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READ ALL ABOUT IT:
DO AUDITORS RESPOND TO PRESS VISIBILITY?

A Dissertation in
Business Administration

by

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ABSTRACT

I examine whether auditors respond to business press coverage. Using the number of articles covering a client firm (a proxy for *visibility*) I find multiple forms of evidence that auditors do respond to the *visibility* of their clients. I first find that auditors charge higher audit fees for clients with higher *visibility* incremental to other determinants of fees. These increased fees are consistent with predicted increases in auditor effort and perceived auditor business risk. The audit fee evidence is corroborated using other proxies for auditor decisions: audit report modifications, the level of accruals, and auditor turnover. The evidence suggests that auditors respond to risks driven by press coverage. Additionally, I find evidence of costs of media coverage to auditors. First, I find that the propensity for a client firm to receive an AAER from the SEC is increasing in the prior year's *visibility*. Second, I find that the total revenue an auditor receives from its entire client portfolio is decreasing with the absolute dollar value of restatements by high *visibility* client firms. These findings indicate that auditors must consider both reputation and monetary concerns in audits of clients subject to high media *visibility*. I conclude that the press not only participates in the external monitoring of financial reporting, but that through its influence on official external monitors the press actually also affects financial reporting.

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DEDICATION

For Mikah.

You never gave up on me, and you never let me give up.

1. INTRODUCTION

Members of the Society of Professional Journalists believe that **public enlightenment is the forerunner of justice** and the foundation of democracy. The duty of the journalist is to further those ends by seeking truth and providing a fair and comprehensive account of events and issues. Conscientious journalists **from all media and specialties** strive to serve the public with thoroughness and honesty. Professional integrity is the cornerstone of a journalist's credibility. Members of the Society share a dedication to ethical behavior and adopt this code to declare the Society's principles and standards of practice.

—*Preamble to the Society of Professional Journalists Code of Ethics* (emphasis added)

One of journalists' fundamental roles is that of a gatekeeper—responsible for selecting which subjects and stories receive circulation in the press. Journalists perform this gatekeeping role by both disseminating public information and uncovering private information. Recent research provides evidence of different ways that the press identifies and disseminates important business news which include highlighting excessive executive compensation (Core, Guay, and Larcker 2008), exposing board ineffectiveness (Joe, Louis, and Robinson 2009), and identifying fraud (Miller 2006; Dyck, Morse, and Zingales 2010).¹ In this paper, I study one effect of the gatekeeping role played by the business press—that the volume of press coverage targeting individual companies increases the incentives of official watchdogs to monitor financial reporting. Specifically, I ask whether auditors respond to the *visibility* of their clients in the business press.

I provide evidence that measures of business press coverage² are associated with

¹ One reason for auditors to respond to business press coverage of their clients is the concern that the press can uncover fraud that was not prevented or detected during the preparation of the financial statements or during the external audit. Auditors have incentives to avoid the reputational costs of the business press uncovering such problems that investors expect auditors to identify. However, due to the low frequency of fraud this is not considered to be the auditors' primary concern pertaining to business press coverage.

² I use the terms *visibility*, media coverage, and business press interchangeably throughout the paper. Variable definitions for the media coverage variables are detailed in Appendix A.

several auditor decisions. I use auditor effort³ as the primary measure of auditor response to press coverage. I also perform tests using measures of audit opinion modifications, reported accruals, and auditor turnover.⁴ I find that audit fees, a proxy for effort, are larger for companies that are more visible in the media even after controlling for other known fee determinants. I also report a positive association between business press coverage and an auditor's propensity to include modified audit opinion reporting language. Tests of the propensity to issue going concern opinions yield similar results. I also find that signed accruals (a proxy for reporting quality) are negatively related to press coverage. Finally, I find that the probability of auditor turnover is positively associated with press coverage. The evidence suggests that auditors do respond to business press coverage of their client firms.

As support for my predictions that auditors will respond to increased media coverage of their client firms, I provide evidence of both reputational and monetary costs auditors experience in association with their highly visible clients. I first document that audit client *visibility* is positively associated with the probability that the Securities and Exchange Commission (SEC) will issue an Accounting and Auditing Enforcement Release (AAER) for the client in the next year. This finding is consistent with the notion that the SEC uses the press as a guide in allocating its limited resources to investigate

³ I use audit fees as a proxy for auditor effort. While an alternate measure such as the number of hours worked would be preferable, I rely on the assumption that fees are strongly correlated with audit hours. Interviews with current and former auditors with levels of experience ranging from partner to new hires support the viewpoint that audit fee estimation is hour driven.

⁴ Francis and Krishnan (1999) summarize five tactics auditors select from when auditing firms with large accruals. They use "effort" specifically to denote increasing audit procedures, but the other choices are not likely to be effortless. Alternative actions including lowering the threshold for issuing modified reports, requiring accrual reducing adjustments, and charging increased risk premiums will all require justification and negotiation with clients. Even filtering out the client would call for the replacement auditor to consider the effect of media coverage on engagement risk. I consider effort, opinion modifications, reported accruals, and turnover each separately as evidence that auditors do respond to press *visibility*, but acknowledge that the primary decision I investigate, auditor effort, is likely related to many of the proxies I employ.

firms. The implication that an AAER has for the auditor is a reputational cost.

The second cost I document which auditors face related to client *visibility* in the business press is the effect of having high *visibility* restatements on their audit client portfolio. I show that the absolute value of restatements by firms in the top *visibility* quintile are negatively associated with an auditor's total fee revenue in the subsequent year. Together with the AAER evidence, this indicates that the reputational costs experienced by auditors can have real monetary consequences motivating auditors to assess and respond to client *visibility*.

The reported findings have implications for multiple parties. Auditors can be informed regarding the reputational and monetary costs associated with the *visibility* of their clients. Client firms should be aware of the additional scrutiny by auditors and the SEC associated with their own *visibility*. When making decisions to influence their prominence in the media, client firms should take into account the result that coverage of all tones is associated with increased levels of auditor effort. Policy makers and enforcement authorities should consider both how auditors respond to firm *visibility* and how media coverage is associated with their enforcement efforts. Similarly, the business press should recognize its potentially influential role in the external monitoring of financial reporting. Finally, these results have implications for auditing standards and may need to be addressed by standard setters.

2. BACKGROUND AND MOTIVATION

Financial reporting is central to the efficient functioning of capital markets and as such there has long been strong concern for improving reporting and disclosure (Basu and Waymire 2006; Barton and Waymire 2004). One critical aspect of improving reporting and disclosure is external auditing (Watts and Zimmerman 1983; Kinney 2005). The role of auditing has come under increased scrutiny in the last decade. For example, in response to a series of major accounting scandals, the passage of the Sarbanes-Oxley Act of 2002 included several sections affecting auditors.⁵ More recently the Public Company Accounting Oversight Board has reconsidered implementing auditor rotation requirements to improve auditor independence (PCAOB News Release August 2011) and has solicited public comments on proposed changes to the auditor's reporting model.⁶ However, auditing as an effort to provide assurance on and improve financial reporting disclosure can hardly be considered in isolation (Beyer, Cohen, Lys, and Walther 2010; Berger 2011). Accordingly, this study analyzes the effect of the business press (a key component of the information environment) on the effort and effectiveness of the institutional monitoring (by auditors) of financial reporting.

⁵*Title I* for example details the creation of the Public Company Accounting Oversight Board which ended the self-regulation of the audit industry.

⁶ Proposed changes under consideration include: the addition of an auditor's discussion and analysis, required and expanded use of emphasis paragraphs, auditor assurance on other information outside the financial statements, and clarification of language in the standard auditor's report (PCAOB News Release June 2011)

2.1 The Business Press and the Auditor's Risk Environment

There are at least two reasons to expect the business press to influence auditors' decisions. First, press coverage influences auditor business risk.⁷ In a footnote of Statement on Auditing Standards 47 auditor business risk is described as follows: "In addition to audit risk, the auditor is also exposed to loss or injury to his professional practice from litigation, adverse publicity, or other events arising in connection with financial statements that he has examined and reported on."⁸ The risk of adverse publicity from connection with a client is greater for clients that are already receiving more press.⁹

The reputation effect of adverse publicity is perhaps the most important effect of the media on auditor decisions and is highlighted by the demise of Arthur Andersen. Ball (2009) summarizes some of the research on the reputational damage of the Enron scandal and conjectures that even without regulatory intervention "the audit market would have closed Andersen on its own accord, because the firm's greatest asset (a reputation for quality, independent auditing) . . . [was] in ruins." In related work, Barton (2005) shows that client firms with high media visibility were more concerned about auditor reputation and responded to the Enron scandal by replacing Arthur Andersen as their auditor sooner than less visible firms.¹⁰ While the Enron case is an extreme example of fraud and audit

⁷ Auditor business risk is distinct from audit risk, the latter of the two being defined as a function of inherent risk, control risk, and detection risk.

⁸ This definition of auditor business risk is consistent with common audit textbook definitions of engagement risk (Arens, Elder, and Beasley 2010; Messier, Glover, and Prawitt 2006). Auditor business risk is also defined as a component of engagement risk closely related to client business risk (Colbert, Luehlfigg, and Alderman 1996). Arens, Elder, and Beasley (2010) acknowledge that there is some disagreement regarding the ability and appropriateness of auditors attempting to reduce auditor business risk through methods other than the client acceptance/retention decision.

⁹ Green, Hand, and Penn (2012) find that coverage in the previous month is a strong predictor of the dissemination a news story will receive. They also find that bad news bias (defined as the wider dissemination of bad news stories relative to good news and neutral stories) is greater for firms that were subject to high levels of press coverage in the prior month.

¹⁰ Barton (2005) measures press visibility as the number of articles a firm appears in during the year preceding his sample period where the client is named in the headline or first paragraph of the article.

failure, Barton (2005) and the research summarized in Ball (2009) support the significant role of the business press in increasing auditors' exposure to severe declines in reputation.

The example of Arthur Andersen emphasizes the potential costs of damaged reputation following a serious audit failure. Auditors also suffer reputational damage and costly litigation in cases when audits are performed in compliance with standards as proposed in SAS 47. By design audits provide reasonable rather than complete assurance, so there is always some risk that auditors will fail to detect and report on material misstatements. The probability of investors, regulators, or the media uncovering material misstatements that have been missed by auditors increases as the media and other stakeholders more closely scrutinize financial reports.^{11,12} The heightened scrutiny spurred on by media coverage increases the expectation that fraud and material misstatements from GAAP that auditors fail to detect will be subsequently uncovered increasing both litigation costs¹³ and negative reputation effects for the auditor.¹⁴

Auditors also face the concern of increased scrutiny by regulators such as the SEC who may target investigations for client firms that are in the business press' spotlight. SEC investigations increase the probability that undetected material misstatements are

Article tone is not assessed.

¹¹ Anecdotal, two "Grumpy Old Accountants" have tried to help by frequently writing on deviations from GAAP by Groupon (a company subject to substantial press coverage). In a blog entry on September 24, 2011, they express satisfaction that their writing has had an "impact on practice" noting that they had sent a copy of a previous article to the SEC and reported on improper revenue recognition practices through the SEC's Whistleblower Program. Groupon has since restated revenues. See: <http://blogs.smeal.psu.edu/grumpyoldaccountants/archives/327>.

¹² Increased media coverage can increase the probability that misstatements are detected even if the media itself plays no role in discovery. If coverage brings a company to the attention of other parties such as the SEC or other financial experts and they uncover errors or irregularities, then the effect of increased media coverage is still a legitimate risk to consider.

¹³ Even when audits have been properly executed following the standards of the Public Company Accounting Oversight Board (United States), auditors bear the costs of defense or settlement when sued.

¹⁴ Similar to the discussion of fraud in footnote 1, I do not expect the uncovering of material misstatements by the press to happen frequently enough to be the primary risk factor of concern to auditors. I do recognize it as a possible source of risk though. Since I do not attempt to separate the different sources of risk related to *visibility*, I identify each acknowledging that some may be deemed to have a negligible effect.

uncovered. I expect regulators consider media coverage in their selection process because they have limited resources and can increase the visibility of their efforts by targeting highly visible firms for investigations and regulatory actions. This selection strategy also reduces the set of companies investigated to firms in which the investing public is perceived to have the greatest interest.¹⁵

A second reason to predict that auditors respond to the business press is because journalists can uncover private information such that auditors learn details about management incentives, the corporate environment, or the probability of reporting irregularities which the client might otherwise hide from its auditors. However, because auditors have specialized technical training in evaluating financial statements, access to private information unavailable to the press, and potentially many years of experience auditing the firm's financial statements, coverage by the press may not be informative to the audit process.¹⁶ Unlike the press, the work of auditors is regulated. Auditors follow the standards of the Public Company Accounting Oversight Board (United States) which they need to consistently apply across audits of the specific set of companies they are engaged to report on. Press coverage on the other hand is substantially less restricted in both the method of coverage and the selection of topic. The press is not required to cover all companies equally (or at all) and can report on issues outside of the scope of the financial statement audit. Also, while the press may adhere to codes supporting truthfulness, accuracy, and objectivity, they are not required to follow a set of standards in

¹⁵ Determining the extent to which press coverage influences the SEC's resource allocation is not within the scope of this paper. I do document a positive association between SEC issued AAER releases and a company's prior year *visibility* though.

¹⁶ In discussions with auditors, they do not dispute that they follow the press coverage of their clients and related industries. One partner described the concern that failure to consider risks identifiable through the press would be a great embarrassment.

evaluating the companies subject to their reporting. This lack of standardized investigation will result in coverage that varies in comprehensiveness from story to story. Consequently, the auditor having technical expertise, access to private information, client-specific experience, and extensive performance standards may not be influenced by the type or extent of coverage an auditee receives in the business press.

Even when allowing for the press to play an informative role, it may be the case that the auditor is able to use press coverage to enhance its understanding of a business without responding by increasing audit effort, altering opinion modification thresholds, or altering other decision processes. Auditors can read from the business press to enhance their understanding of the client's industry or business without responding to the clients' relative levels of coverage by revising audit decisions.

An auditor may also desire the prestige of being associated with a highly visible client and respond in ways inconsistent with my predictions in order to acquire and maintain the client. Lower fees, a reduced propensity to modify audit reports, and laxness in permitting larger positive accruals could each result from this strategy. Still, I predict the reputation and litigation risks associated with high *visibility* will lead to auditor responses to press coverage that are consistent with greater effort and increased auditor scrutiny.

These reasons point to the expectation that greater media coverage increases auditors' awareness of or exposure to risk. It is noteworthy that it is not necessary for the press to provide new information to the auditor for this risk adjustment to take place. *Visibility* can increase auditor effort even when news coverage is not investigatory. For that reason, I expect *visibility* to induce an auditor response even when the articles are

primarily dissemination and not investigation oriented. Similarly, the articles could all be favorable towards the company and still affect auditors' decisions simply because the increased positive coverage also represents amplified risk from the additional scrutiny. With the increased positive coverage the risk of adverse publicity is also heightened for those clients that subsequently experience a negative event.

To be more precise, the most visible firms represent the highest auditor business risk because should the client fail or should some other undesirable outcome occur, the risk of adverse publicity for the auditor is greatest. Therefore, *visibility* does not represent past or current adverse outcomes so much as risk of adverse publicity should adverse outcomes occur in the future.

2.2 Auditor Responses to Press Coverage Risks

Auditors choose from multiple actions to reduce exposure to relevant risks. Choices include increasing auditing effort, modifying audit opinions, negotiating lower accruals, or dropping a client (Krishnan and Krishnan 1997; Francis and Krishnan 1999; Schelleman and Knechel 2010). For a given level of risk tolerance, auditors can reduce overall audit risk by increasing the extent of their auditing procedures.¹⁷ This method of risk reduction leads to more thorough and more costly audits which correspond to greater audit fees.

Although I argue for a positive relation between *visibility* and auditor effort, this

¹⁷ There are both philosophical and practical arguments against auditors attempting to offset auditor business risk by reducing audit risk. Ideally, clients engage and compensate an auditor to provide reasonable assurance that financial statements are not materially misstated from GAAP and should not be required to purchase a more expensive and more thorough audit to offset and/or reduce the expected costs of auditor business risk. Bell, Landsman, and Shackelford (2001) report that in their discussions with practitioners multiple reasons were given that auditor business risk should not or could not be priced into the audit. Bell, Landsman, and Shackelford (2001) find however that audit hours billed for an engagement are increasing in the perceived auditor business risk for a large international audit firm.

relation is not obvious. An alternative prediction is that auditors may perceive effort reducing benefits to client media coverage. Since the media can serve as an additional unofficial monitor, the auditor might anticipate that auditees subject to greater coverage will proactively choose to report more accurately. This would be consistent with the findings of Joe, Louis, and Robinson (2009) who show that media exposure of board ineffectiveness leads to improvements in corporate governance. I expect that despite this potential improvement, auditors will be skeptical. They may be concerned that their client firms will not improve rapidly enough to consider media coverage to be risk reducing. This suggests that a study on the long term effects of media coverage may merit further investigation. Regarding the setting examined in Joe, Louis, and Robinson (2009), the board changes may also result in the auditor receiving more support to work harder as well (Abbott, Parker, Peters, and Raghunandan 2003). As a result, media scrutiny can lead to creating a board that will encourage greater auditor effort.

Another reason to expect a negative association between *visibility* and auditor effort is that auditors may perceive a benefit to being associated with clients that are highly covered in the business press. Auditors might seek to obtain the perceived prestige of having clients with high *visibility*. In order to attract and retain high *visibility* clients, auditors could make choices which would appear to indicate effort reductions. As an example auditors may offer fee discounts or alter thresholds for issuing modified opinions. I predict that despite this conflicting incentive, auditors will not disregard the potential costs of being associated with a high *visibility* client and will increase audit effort accordingly.

Following the argument that media coverage leads to greater auditor effort my

primary tests are aimed to address the hypothesis stated in the alternative form:

H1: Audit effort is positively associated with the client firm's media coverage *visibility*.

I use the proxy audit fees to test auditor effort. I also consider other auditor responses to business press coverage which are related to the effort decision.

Another choice auditors can make is to increase their propensity to issue modified audit reports including issuing going concern opinions. This decision results in a lower threshold for opinion modification and is indicative of greater auditor independence from the client firm. This auditor decision is aimed at protecting the auditor's reputation which would be damaged if the auditor issued a clean opinion and a client firm subsequently was discovered to have merited an opinion modification.¹⁸ The related hypothesis is stated in the alternative:

H2: The probability that an auditor issues a modified audit opinion is positively associated with the client firm's media coverage *visibility*.

In testing H2 I consider both a broad measure of opinion modifications as well as the specific inclusion of a going concern opinion modification.

Auditors may also respond to press coverage by choosing to be less tolerant of aggressive accounting estimates. Observing a negative relationship between signed accruals and *visibility* would provide evidence of this auditor response. This relationship

¹⁸ Although the opinion modification can protect the auditors reputation under the presumption that the state of the client will subsequently be revealed, the auditor may also consider the opinion modification to be detrimental to their present reputation and relationship with the client. This could lead to predictions that more visible firms will be less likely to modify the audit opinion. While I acknowledge this alternate possibility, I base my predictions on the assumption that when auditors deem an opinion modification to be necessary, they assess the benefits of reporting it to be greater for highly visible clients.

results from auditors placing greater scrutiny on potential overstatements than understatements (Barron, Pratt, and Stice 2001). Positive accruals can be determined to be overstated and negative accruals can be assessed to be insufficiently low and be adjusted further downward. Negotiating lower accruals is a strategy auditors can use to counter the uncertainty of accruals and is expected to increase with the risk associated with *visibility*. The hypothesis is stated in the alternative as follows:

H3: Signed accruals¹⁹ are negatively associated with the client firm's media coverage *visibility*.

A final auditor response to consider is that the auditor could drop the client. If the auditor considers the risk associated with press coverage to be too great to offset through the decisions already discussed, the auditor may decide to end the audit engagement.

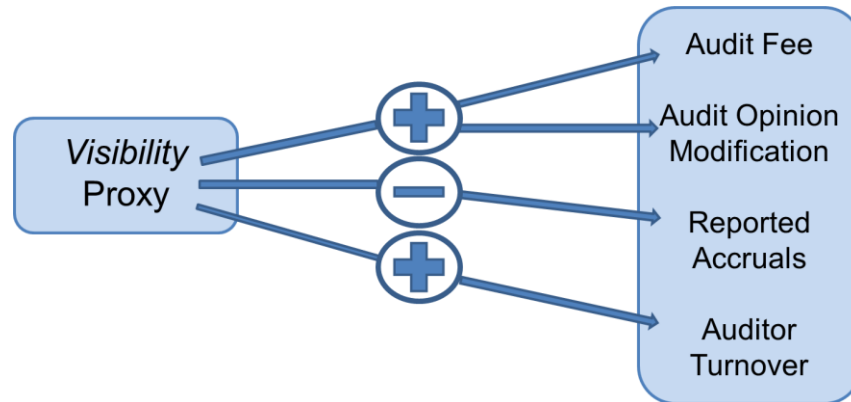
Stated in the alternative, the related hypothesis is as follows:

H4: The probability of auditor turnover is positively associated with the client firm's media coverage *visibility*.

The predicted auditor responses to media coverage of audit clients are summarized below in Figure 1 with the hypothesized coefficient signs encircled:

¹⁹ While absolute accruals have been used to assess audit quality, they would not be appropriate for testing this hypothesis. This prediction is directional and assumes an asymmetric auditor response to media coverage. The expectation is that the auditor will respond to media coverage by attempting to reduce income increasing accruals. The auditor does not have complete control over reported accruals, but it does have the ability to withhold a clean opinion if they are perceived to be materially misstated.

FIGURE 1 – Summary of Predicted Auditor Responses to Client Media *Visibility*



2.3 Press Coverage Proxies

To measure attributes of media coverage, I use news story analytics from RavenPack.²⁰ To capture a company's level of *visibility* in the business press, I use the number of stories in which the company appeared in the headline or was identified as having a significant role in the story²¹ (e.g. the company is the plaintiff or defendant in a story about a pending lawsuit).²²

RavenPack also includes sentiment analytics on each story. Using multiple specialized business event focused dictionaries, stories are assigned positive, neutral, or negative sentiment scores.²³ To alleviate the concern that results are driven by underlying

²⁰ RavenPack is a provider of real time news analytics whose primary clientele include large and second tier investment banks. Information regarding their current products can be found at: <http://ravenpack.com/>.

²¹ The possible roles identified are listed in Appendix B as categories which are separated under various news story group headings. This appendix as well as Appendix C and Appendix D contain similar content to Green, Hand, and Penn (2012). The content was obtained from RavenPack reference materials as well as from company representatives.

²² It is possible that a company will not be named in the headline, but will still be identifiable as having a significant role in the story. Related to the litigation example, if a headline reads “Company ABC Faces Lawsuit” and the details of the story indicate Company XYZ is the plaintiff, then Company XYZ will be identified as having a significant role and will receive a high relevance score in the article as described in Appendix C.

²³ The news story analytics are described in Appendix C. Examples of news story dissemination patterns are provided in Appendix D. Correlations between the measures of news sentiment are provided in Appendix E. They are significantly positively correlated indicating general agreement in their assessments

bad news events rather than actual press coverage attributes, I also create *visibility* scores partitioned by news story tone. In additional tests I measure the mean tone of these articles to capture the company's *prevalent sentiment* in the business press.

The results in this paper emphasize the importance of understanding the information environment (Beyer, Cohen, Lys, and Walther 2010; Berger 2010) as a whole when determining the effectiveness of the monitoring of financial reporting. I find evidence that suggests that the business media enhances the monitoring role of at least one key monitor, auditors. I find that auditor effort increases in response to press coverage and that auditors also respond to media coverage through opinion modifications, accrual adjustments, and auditor turnover. These auditor responses are consistent with the potential associated reputational and monetary costs I identify.

2.4 Examples of Auditor Costs of Clients' Press Coverage

While the focus of this study is to examine how auditors respond to their clients' business press coverage, I also provide evidence of the costs auditors face related to press coverage. I document the positive relationship between press coverage and the probability that a client will receive an AAER as one reputational cost. I also document the negative relationship between restatements by high *visibility* clients and an auditor's total audit fee revenues received in the next year. I do not provide formal hypotheses here, but do describe the research design and coefficient predictions in Chapter 3.

of news story sentiment.

2.5 Relevant Research

Supplying evidence of a relationship between attributes of press coverage and decisions made by the external auditor is an important step in the growing literature on the role of the business press. Core, Guay, and Larcker (2008) show that the media follows a “sophisticated approach” in story selection evidenced by the relationship between negative coverage and excess compensation as opposed to raw compensation. They do not find evidence that this coverage has any influence on the excessive pay structures or CEO turnover though. The titular implication of these findings is that the press has no power to influence firms’ executive compensation practices.

Recent research provides evidence that the business press does operate as a unique information intermediary influencing firms’ information environments as the press both disseminates previously disclosed information and generates new information (Bushee, Core, Guay, and Hamm 2010). The dissemination of firm information by the business press has been shown to impact trading and other firm-level market activity measures including bid-ask spreads and idiosyncratic volatility (Engelberg and Parsons 2011; Soltes 2010). There is also evidence that the sentiment of press coverage relates predictably to market prices and trading volume (Tetlock 2007). Previously noted evidence suggests that in generating new information the press plays an external monitoring role as a watchdog identifying a large proportion of discovered instances of accounting fraud prior to monitors that are actually tasked to detect and uncover fraud (Miller 2006; Dyck, Morse, and Zingales 2010).²⁴

In addition to these studies which have examined the role of the press and its

²⁴ As discussed in footnote 1. I do not expect the “watchdog” role of the press to be the primary cause of auditor responses due to the infrequent nature of these events.

influence in the market place, Kothari, Li, and Short (2009) provide some initial evidence of the interaction between press coverage and other information intermediaries.

Dependent on coverage sentiment, they show that business press coverage affects firm-specific measures of the cost of capital, stock return volatility, and analyst forecast dispersion. The effect of press coverage attributes on analyst forecast dispersion is an interesting first look at how the business press as one information intermediary influences another intermediary. The business press/analyst setting does not, however, incorporate the differential access to private information that exists²⁵ between the business press and auditors since analysts are no longer privy to such information in the post Reg FD environment.

Some research has also investigated the relationship between bad news press coverage and audit opinions. In an experimental setting Joe (2002) finds that auditors are more likely to issue going concern opinions if they are presented with a bad news event article along with the auditee's financial statements.²⁶ Using the experimental setting, Joe (2002) is able to design the news article so that it does not convey information to the auditor that is not already contained in the financial statements.²⁷ Her findings are consistent with archival studies by Frost (1991) and Mutchler, Hopwood, and McKeown (1997) which both find that auditor assessments are influenced by media coverage of bad

²⁵ It is important to recognize that while auditors do have greater access to private information necessary for the performance of their attestation duties, this does not mean that they have free access to all firm records. Since requests for information are made to the client firm, the firm does have an opportunity to attempt to hide information from or misreport information to the auditor. This is one of the challenges that make fraud more difficult to discover than a simple accounting error.

²⁶ Specifically, Joe (2002) considers loan default by the audit client.

²⁷ Because experiment subjects were only asked to assess a single set of financial statements, it is uncertain whether those that received an accompanying news article viewed its inclusion as a representation of the client's *visibility*. It is possible that they could have assumed that receiving just one article was indicative of a low *visibility* client, but were still affected by the salience effect Joe (2002) examines.

news.²⁸ These prior studies on the relationship between press coverage and auditing have focused on specific bad news events and do not consider the full media coverage environment of the client or the effects of media coverage of good news events.²⁹

²⁸ Frost (1991) examines whether new loss contingencies receive financial statement disclosure and result in qualified audit opinions. Mutchler, Hopwood, and McKeown (1997) find that auditors are more likely to issue a modified audit opinion for debt defaults by audit clients when the defaults receive coverage in *The Wall Street Journal*.

²⁹ There have also been studies which consider the relationship between the press and auditor effort in alternate settings. In a qualitative field study of the U.S. General Accounting Office's (GAO) reporting process, Basu, Dirmsmith, and Gupta (1999) note that all GAO members participating in the study acknowledged an awareness of the press coverage some audits received and admitted to altering audit efforts for such audits, especially if the coverage had been critical. Similarly, Redmayne, Bradbury, and Cahan (2010) examine the relationship between political visibility and audit effort using a small sample of New Zealand public sector companies which follow the same accounting standards as public companies in New Zealand, but are owned by the government and thus face the ongoing concern of potential political intervention.

3. METHODOLOGY

The primary challenges to my study include the imperfect measurement of key variables of interest and the potential for omitted correlated variables. Audit fees are the proxy for the main decision I investigate, auditor effort. Some problems with this measure include that a portion of it may be attributable to a risk premium and there is no way to detect what portion is actually due to increased work.³⁰ Ideally I would like to directly observe auditor effort in response to external stimuli and the audit fee is an imperfect proxy. However, recent research provides some assurance related to this proxy (Hribar, Kravet, and Wilson 2010).³¹ To further mitigate this concern I perform additional tests of other auditor decisions which also have implications regarding auditor effort.

Another concern is that the variables for media coverage attributes may not be measured over the correct time period. To the extent that the measures capture “old news” which the auditor does not deem relevant in their risk assessment or indicative of future coverage, the effects may be dampened. Finally, regarding potential omitted correlated variables, it is possible that media coverage attributes do not affect the audit process, but that the underlying news events are driving the results. To alleviate this concern I control for firm performance and the amount and tone of news that a firm experiences. I include market adjusted cumulative returns as a control for the average tone of events

³⁰ Bell, Landsman, and Shackelford (2001) are able to directly examine the number of hours billed and the billing rate per hour charged for a sample of engagements of one large auditor. One of their findings is that the amount of audit hours, not the hourly fee rate, is increased when auditor business risk is assessed to be high. If this result is applicable to other auditors, it supports the use of increased fees as a proxy for increased audit hours billed on an engagement. A lingering concern is that even with access to the actual number of hours billed, unbilled hours are not known. I assume that auditor hours and auditor fees are correlated.

³¹ Specifically they find that firms with larger residuals from an audit fee model are lower quality. They posit that unexplained audit fees are larger because auditors recognize the low quality accounting and respond by increasing their effort incremental to what would be expected based on the audit fee model's explanatory variables.

experienced by a firm over the period examined along with measures of return volatility and trading volume to capture the amount of news events that a firm experienced during the year.³² Including these variables requires media coverage attributes to be associated with auditor decisions incrementally to the market responses to all forms of news released during the measurement period in order to find significant results.

3.1 Audit Fees as a Proxy for Auditor Effort

I rely on prior³³ and concurrent literature in selecting audit fee determinants which may be correlated with *visibility*. Hribar, Kravet, and Wilson (2010) develop a comprehensive model of audit fee determinants which I draw from. Consistent with Price, Sharp, and Wood (2010), I omit some variables which would restrict the size of my sample and make adjustments based on data availability. Detailed descriptions of variable calculations are included in Appendix A.

To test H1 I regress my proxy for auditor effort, the natural log of audit fees (*LnFee*), on measures of firm *visibility* estimated over different time windows corresponding to the fiscal year audited. As noted earlier, one of the challenges in assessing the influence of media coverage is selecting an appropriate window to calculate the media coverage variables. According to discussions with practicing auditors, the audit fee is typically set sometime from the end of the first fiscal quarter to the end of the

³² These returns measures are not typical of standard audit fee models, but they are included to control for the attributes of the news events underlying the actual news coverage. This is intended to permit inferences on how auditors respond to the press coverage itself and not on the underlying news. It should also be noted that as return volatility increases, the probability that stockholders will experience losses and pursue litigation against the auditor is affected. I thank Kenny Reynolds for this observation.

³³ Audit fee models have been studied extensively. In a survey of the literature, Hay, Knechel, and Wong (2006) summarize the research following the seminal work of Simunic (1980) and identify numerous independent variables that have been tested as determinants of audit fees. Hay, Knechel, and Wong (2006) provide a taxonomy classifying these variables categorically as client attributes, auditor attributes, and engagement attributes.

second quarter. In unusual circumstances, there can be adjustments or changes in scope which lead to increased fees to compensate for unanticipated additional effort.^{34,35} To reflect the more common timing for setting fees, media coverage is estimated over the first two fiscal quarters. Then to allow for scope changes and fee adjustments, measures of media coverage are estimated over the entire fiscal year.

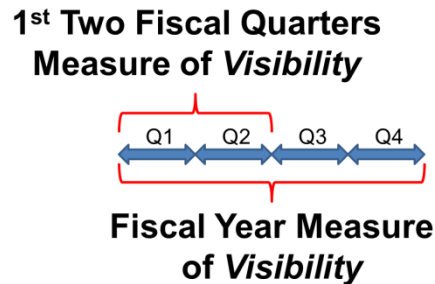
My primary variables of interest are *LnVis_EarlyQuarters* and *LnVis_FiscalYear* which capture a company's level of *visibility* in the business press. *LnVis_EarlyQuarters* is the natural log of one plus the number of stories a firm appears in during the first two quarters of the fiscal year. This is the time period that audit effort is planned and fees are more commonly established. This early quarters measure will only be missing relevant news items for firms that subsequently adjust audit scope. *LnVis_FiscalYear* is calculated over the entire fiscal year. *LnVis_FiscalYear* is also used for tests of the other auditor decisions. Media coverage variables measured during the fiscal year contain information about the news coverage which had been released prior to the beginning of the audit as well as coverage which took place during much of the audit performance. These annual measures of media coverage capture the information set that auditors would most likely be influenced by during the critical planning stages of an audit as well as the information related to the current year's financial statements which could potentially influence auditor decisions.

³⁴ Scope changes were described as generally uncommon, but it was noted that following the market crash of 2008 scope adjustments were widespread due to the unanticipated effort needed to reassess valuations post-crash. A senior manager with a Big 4 auditor qualified this statement saying that changes to the initially agreed upon audit fee can be more frequent depending on the preferences of the managing partner, the audit client, and the circumstances of the audit.

³⁵ Anecdotally, a senior manager for one Big 4 firm reported a recent engagement fee was not agreed upon until after the audit was completed.

A simple depiction the timeline for *visibility* proxy measurement appears in Figure 2 below:

FIGURE 2 – *Visibility* Variable Measurement Timeline



For news printed after the fee is set when there is no adjustment to audit scope, the auditor may also perform additional work without requesting additional compensation in the current year.³⁶ In such cases I would expect the auditor to take the additional effort into account when determining the audit scope and negotiating fees in the following year. For this reason I also examine the association between media coverage attributes and the subsequent year's audit fee. In untabulated results, I show the main findings are robust to this specification.

As mentioned earlier, one of the greatest concerns is that the statistical associations I observe do not represent the relationship between the media and auditor decision proxies, but are instead attributable to an omitted correlated variable pertaining to the actual news events experienced by the firm. It is possible for example that increased effort occurs solely due to distress experienced by firms or other risk increasing

³⁶ A former Big 8 auditor gleefully reminisced about learning the motto shared by the managers in his first office: Work late and charge eight!

events that are strongly correlated with news coverage. I attempt to rule out this threat to my inferences by controlling for several proxies for firm distress and newsworthy events.

To control for the market's assessment of the underlying news events I include the variables *CRet_EQ*, *RetVol_EQ*, and *TradeVol_EQ*, each measured over the course of the first two quarters of the fiscal year. I also calculate corresponding measures for the entire year identified with the suffix *FY*. *CRet_EQ* is a company's cumulative market adjusted return for the year being audited. I include this to account for the market's assessment of a firm's news sentiment so that my news measures will capture the effect of media coverage while controlling for the market's assessment of a firm's actual performance. Similarly, I include *RetVol_EQ* and *TradeVol_EQ* to account for the volume of news events taking place during the year. *RetVol_EQ* is the standard deviation of a company's daily market adjusted returns during the fiscal year. *TradeVol_EQ* is the standard deviation of a company's daily trading volume adjusted for the number of shares outstanding. I include these so that my news measures will reflect the effect of media coverage a company receives after controlling for the level of activity in the market during the year to control for the frequency of potential news events.

I also include controls for newsworthy characteristics of the audit clients. I include *LnAssets* which is the natural log of total assets. I include *ROA* which is a measure of the return on assets, and *Loss* which is an indicator variable equal to one if income before extraordinary items was negative in the current or either of the two previous years. Both of these measures attempt to capture the performance and possible distress a company is experiencing. *Debt* is also included and indicates how severely leveraged a firm is. *ModOpinion* is an indicator variable set equal to one when the audit

report includes anything other than a standard unqualified audit opinion. *Acq* is an indicator variable set equal to one when a firm has non-zero acquisition or restructuring costs. Other controls for firm size and litigation risk are also included.

Collectively these controls are intended to capture the nature of the underlying news events providing some evidence that the primary results are not driven by the nature of the company, the news coverage, or events coinciding with different levels of news coverage. The remaining control variables are defined in Appendix A and are included to control for audit complexity, inherent risk, and litigation risk. With the exception of indicator variables, all variables are winsorized at the 1 and 99 percentile levels. Equation (1) below is run as a panel regression with standard errors clustered by firm.³⁷ Year (*Year*) and industry (*Ind*) fixed effects are also included.

My model for tests of H1 with the natural log of audit fees as a proxy for effort is as follows with subscripts omitted for convenience:

$$\begin{aligned}
 \text{LnFee} = & \beta_0 + \beta_1 \text{LnVis_EarlyQuarters} + \beta_2 \text{Big4} + \beta_3 \text{LnAssets} + \beta_4 \text{LnMVE} \\
 & + \beta_5 \text{Inventory} + \beta_6 \text{Receivables} + \beta_7 \text{Debt} + \beta_8 \text{ROA} + \beta_9 \text{Loss} \\
 & + \beta_{10} \text{ModOpinion} + \beta_{11} \text{LitRisk} + \beta_{12} \text{Acq} + \beta_{13} \text{CRet_FY} + \beta_{14} \text{RetVol_FY} \\
 & + \beta_{15} \text{TradeVol_FY} + \Sigma \text{Year} + \Sigma \text{Ind} + \varepsilon
 \end{aligned} \tag{1}$$

The above analysis identifies associations between audit effort and levels of media coverage variables. As a stricter test of the relationship I also estimate how changes in media coverage attributes relate to changes in audit fees. My predictions are consistent with the discussion for a levels test in that I expect increases in *visibility* to

³⁷ Multiple observations are included for identical firms for which attributes such as firm size and media coverage are sticky over time raising concerns regarding independence across observations. In primary tests I cluster standard errors by firm to alleviate such concerns. Additionally, as a robustness test I restrict the sample to include a single observation for each firm. Despite the reduction in power, the results are qualitatively similar and remain statistically significant.

result in higher audit effort evidenced by increased fees.

3.1.1 Tone Conditional Measures of Visibility

One concern is that the results are attributable to the events underlying the business press coverage and not to the actual coverage attributes. I include several controls described above to control for market responses to all news, major firm events, and firm performance characteristics. I also partition the measure of firm *visibility* based on the tone of the news.³⁸ I create three new *visibility* measures which correspond to good news, bad news, and neutral news story coverage, and I substitute these measures into the audit effort model (1). I replace *LnVis_EarlyQuarters* with the tone based partitions of the news data.

LnBadNewsVis_EQ is the natural log of one plus the number of stories about a client firm with a Composite Sentiment Score (*CSS*) value that indicates negative tone. The *CSS* value is a measure of the sentiment or tone of each news story and is subjected to a linear transformation so that positive values indicate a positive story sentiment, the value zero indicates a neutral sentiment, and negative values correspond to a negative sentiment or tone. The calculation of *CSS* is described in greater detail in Chapter 4.

LnNeutralNews_EQ and *LnGoodNews_EQ* are calculated similarly to *LnBadNewsVis_EQ* for stories where *CSS* indicates a neutral or positive tone respectively.

Including the separate *visibility* measures permits an asymmetric response by auditors to news conditional on the tone of coverage. This design allows me to assess

³⁸ Joe (2002) suggests examining the influence of redundant positive news coverage on auditors as a future area for research.

whether bad news coverage is driving the results in the primary analysis and will also indicate whether good news and neutral news are incrementally effort increasing when controlling for the level of bad news story coverage. I expect each news measure to be positively associated with auditor effort. As discussed above, I assume that one of the major concerns auditors consider to be associated with media coverage is the potential adverse publicity and reputational costs they will be exposed to if a negative event occurs in the future and they are associated with the client. For that reason, higher levels of coverage pertaining to good news has a similar potential cost to neutral or bad news coverage of the client.

3.2 Increased Propensity to Issue Modified Audit Opinions

There are other aspects of auditing work and output that reflect auditor decisions aside from audit fees. I next test the type of audit report issued for evidence that auditors respond to media coverage. Auditors can offer opinions that provide some indication on the extent of misstatements or other problems found during the audit. If auditors decrease their level of acceptable audit risk, then I anticipate that the probability of receiving anything other than a clean unqualified opinion will be increasing in firm *visibility*. I also test separately the probability that the audit report contains a going concern opinion modification.

A benefit of examining the audit opinion is that it helps alleviate the concern that fee increases are caused by auditors charging a risk premium without changing any audit procedures or actions. It also provides evidence that speaks to the effect of the media on auditor independence. Consistent with prior research (DeFond, Raghunandan, and

Subramanyam 2002; and Li 2009) I test the propensity of auditors to issue modified audit opinions,³⁹ and I then test the propensity for auditors to modify their reports by issuing a going concern opinion. I follow prior literature in developing the following logistic model where the dependent variable is an indicator variable set equal to one if the audit report includes an opinion modification (specifically a going concern opinion in one variation of the test). The *visibility* variable is as defined above. Other additional control variables include the natural log of annual revenues (*Sales*), the natural log of the market value of equity (*LnMVE*), the change in long-term debt divided by total assets (*ChgDebt*), an indicator variable set equal to one if the firm had negative operating cash flows in the previous year (*Prnocf*), the number of days between the fiscal year end and the auditor's report signing date (*Delay*), and an indicator variable set equal to one if the firm issues new debt in the following year (*NewDebt*).

$$\begin{aligned} \text{Opinion Variable} = & \beta_0 + \beta_1 \text{LnVis_FiscalYear} + \beta_2 \text{Sales} + \beta_3 \text{ROA} + \beta_4 \text{LnMVE} \\ & + \beta_5 \text{Debt} + \beta_6 \text{ChgDebt} + \beta_7 \text{Loss} + \beta_8 \text{Prnocf} + \beta_9 \text{Big4} + \beta_{10} \text{Delay} \quad (2) \\ & + \beta_{11} \text{NewDebt} + \beta_{12} \text{Cret_FY} + \beta_{13} \text{RetVol_FY} + \beta_{14} \text{TradeVol_FY} \\ & + \Sigma \text{Year} + \varepsilon \end{aligned}$$

I also restrict the sample for this test to companies that are experiencing financial distress⁴⁰ defined as firms that report either negative net income or negative operating cash flows during the current fiscal year (DeFond, Raghunandan, and Subramanyam 2002).

³⁹ Consistent with prior literature, I define a modified audit report as one receiving anything other than a standard unqualified opinion coded as one in Compustat.

⁴⁰ This is consistent with research finding that combining stressed and nonstressed firms is not reflective of the auditor's decision problem (Hopwood, McKeown, and Mutchler 1994).

3.3 Signed Accruals as a Proxy for Increased Auditor Scrutiny of Financials

As another test of the effects of media coverage on auditor decisions, I examine the underlying financial reporting that the auditor is opining on. Prior research has used accruals as a proxy for audit quality (see Francis 2004 for a brief review; Venkataraman, Weber, and Willenborg 2008). Conceptually, since accruals are subjective measures, reasonable assurance is challenging to obtain. Also, since manipulating accruals is viewed as a ready means of committing earnings management, auditors should be relatively more concerned when assessing them. Francis and Krishnan (1999) emphasize the risk posed by accruals to auditors because of their uncertainty and the challenges in verifying them even with additional effort.⁴¹ It is presumed that a high quality audit will constrain earnings management and will be associated with smaller absolute accruals. However, prior research suggests that auditors are asymmetric in their responses to the type of problems encountered in their auditing process. Specifically, auditors appear to be more averse to the risk of errors that overstate performance than to the risk of potential performance understatements (Barron, Pratt, and Stice 2001; Abbot, Parker, and Peters 2006). Therefore, I study the relation between *visibility* and signed accruals and predict that they will be negatively related.⁴²

To test the relationship between media coverage and accruals, I run a panel regression using signed total accruals (*Acc*) as the dependent variable with standard errors clustered by firm. I remove observations from the financial industry and observations

⁴¹ In a panel discussion at the 2011 AAA Annual Meeting, Zoe-Vonna Palmrose spoke extensively about using abnormal accruals as a proxy for non-GAAP earnings management. The tenor of her remarks was that abnormal accruals are a very poor measure of material non-GAAP misstatements. Without disagreeing with her remarks, I propose to test accruals because of the risk associated with their uncertainty, and rely on there being a correlation between changes in accruals and auditor efforts to negotiate lower accruals. I do not contend that these changes are indicative of quality though.

⁴² As stated in footnote 19, absolute accruals would not be appropriate to test my predictions.

with extreme absolute accruals.⁴³ Control variables not previously defined include the lagged value of net property, plant and equipment scaled by lagged total assets (*PPE*), the change in total assets scaled by lagged total assets (*AssetGrowth*), the change in total revenue scaled by lagged total assets (*DSales*), net operating cash flow scaled by lagged total assets (*CFO*), and the market value of equity divided by the book value of common equity (*MB*).

$$\begin{aligned}
 Acc = & \beta_0 + \beta_1 LnVis_FiscalYear + \beta_2 LnAssets + \beta_3 LnMVE + \beta_4 PPE + \beta_5 AssetGrowth \\
 & + \beta_6 DSales + \beta_7 ROA + \beta_8 CFO + \beta_9 MB + \beta_{10} Debt + \beta_{11} Loss + \beta_{12} Cret_FY \\
 & + \beta_{13} RetVol_FY + \beta_{14} TradeVol_FY + \Sigma Year + \Sigma Ind + \varepsilon
 \end{aligned} \tag{3}$$

3.4 Increased Probability of Auditor Turnover

Finally, I test the effect of media coverage on auditor turnover. Francis and Krishnan (1999) identify the screening of clients as one of the auditor's potential responses to the uncertainty of accruals.⁴⁴ Krishnan and Krishnan (1997) also examine auditor resignations as a response to litigation risk. Following my prediction that auditors consider increased media coverage of clients to represent increased risk, I expect media coverage to be positively associated with auditor turnover.

The logistic regression model (4) below describes the structure of this test. The dependent variable (*Turnover*) is an indicator variable set equal to one if there is a change of auditors in the subsequent year, and zero otherwise. Control variables not previously defined include the change in sales scaled by lagged sales (*Growth*), an indicator variable

⁴³ I exclude observations where absolute total accruals are greater than the value of total assets. The results are not sensitive to this exclusion.

⁴⁴ Client screening involves both ending existing client relationships and declining to accept new clients. I am only able to observe instances of the former type of client screening.

set equal to one if an auditor has worked for the client for less than four years (*ShortTenure*), and an indicator variable set equal to one if the auditor has worked for the client for more than eight years (*LongTenure*).

$$\begin{aligned} \text{Turnover} = & \beta_0 + \beta_1 \text{LnVis_FiscalYear} + \beta_2 \text{LnAssets} + \beta_3 \text{LnMVE} + \beta_4 \text{Sales} + \beta_5 \text{ROA} \\ & + \beta_6 \text{Growth} + \beta_7 \text{Debt} + \beta_8 \text{Big4} + \beta_9 \text{ShortTenure} + \beta_{10} \text{LongTenure} \\ & + \beta_{11} \text{Cret_FY} + \beta_{12} \text{RetVol_FY} + \beta_{13} \text{TradeVol_FY} + \Sigma \text{Year} + \varepsilon \end{aligned} \quad (4)$$

3.5 Additional Analyses

As an alternate assessment of the tone of coverage, I also consider the effect of a firm's *prevalent sentiment* on auditor effort when controlling for total *visibility*.

PrevSent_EQ is the mean *CSS* value for all stories a firm appears in during the first two quarters of the fiscal year. *PrevSent_EQ* is added to model (1) as an additional explanatory variable. I anticipate that a negative prevailing sentiment in news coverage of a client may be perceived as risk increasing by the auditor even after controlling for the occurrence of the negative events being covered. I therefore predict a negative relationship between *prevalent sentiment* and auditor effort. This would imply that as news coverage becomes increasingly negative, auditors respond by increasing effort.

As a final auditor response test I examine year to year changes to better infer a causal relationship between the media coverage variables and auditor effort proxies. I perform a changes analysis on my primary test of audit effort using audit fees as a proxy as well as a changes analysis when *visibility* is disaggregated by tone.

3.6 Auditor Costs of Client Media Coverage

To support the predictions regarding auditor responses to client *visibility*, it is necessary to understand the costs auditors consider in relation to media coverage. Auditor turnover, one of the predicted auditor responses, results in the loss of an income stream to the auditor. I also propose two additional tests to provide evidence of circumstances where a client's high *visibility* can be costly to the auditor.

In the first test I examine the association between client *visibility* and the probability that the SEC releases an AAER in the following year. I follow Price, Sharp, and Wood (2010) but include *LnVis_FiscalYear* as a new independent variable to assess the impact of media coverage on the probability of receiving an AAER. The dependent variable *AAER* is an indicator variable set equal to one if an AAER is released the next year. I also include *AbsAcc* which is equal to the absolute value of accruals to control for the effect of a firm's reported accruals. Other variables are as previously defined.

$$AAER = \beta_0 + \beta_1 LnVis_FiscalYear + \beta_2 LnAssets + \beta_3 LnMVE + \beta_4 ROA + \beta_5 Debt + \beta_6 Big4 + \beta_7 MB + \beta_8 Loss + \beta_9 AbsAcc + \Sigma Year + \Sigma Ind + \varepsilon \quad (5)^{45}$$

An auditor's reputation is expected to be damaged when one of its clients receives an AAER. When a high *visibility* client receives an AAER, the reputational damage is expected to be exacerbated due to the increased publicity the client is subject to. Quantifying the reputational cost of a client receiving an AAER is beyond the scope of this paper; however, the second cost I examine has direct monetary implications.

In the second test I examine the association between restatements for high *visibility* clients and the total audit fee revenues collected by an auditor in the next year

⁴⁵ Model (5) is tested as a linear probability model. Similar inferences are made from using a logistic regression model.

from the auditor's complete client portfolio. I evaluate model (6) to provide evidence that there is a negative relationship between restatements by high *visibility* clients and an auditor's total audit fee revenue collected from all clients. The dependent variable *LnFeeRev* is equal to the natural log of total audit fee revenue collected by an auditor in year t . The variable of interest, *LnHighVisRestate*, is equal to the natural log of the sum of the absolute value of net restatement amounts of all high *visibility* clients in year $t-1$. High *visibility* clients are considered to be those in the highest *visibility* quintile in year $t-2$. Total audit fee revenues are expected to be negatively associated with *LnHighVisRestate*.

Other control variables include *LnLowVisRestate* which is calculated similarly to *LnHighVisRestate* but is based on restatements for the lowest quintile firms in year $t-2$. *LnAuditorVis* is the natural log of the sum of all news stories about an auditor's clients in the fiscal year and is expected to be positively related to the dependent variable. *LnLagFeeRev* is the natural log of total audit fee revenue received by an auditor in the previous year and is expected to have a positive coefficient. *Big4* is as previously defined and is predicted to have a positive coefficient.

This test of high *visibility* restatement costs is run as a panel regression with year fixed effects and standard errors clustered by auditor. It appears below as model (6):

$$\begin{aligned} LnFeeRev = & \beta_0 + \beta_1 LnHighVisRestate + \beta_2 LnLowVisRestate + \beta_3 LnAuditorVis \\ & + \beta_4 LnLagFeeRev + \beta_5 Big4 + \Sigma Year + \varepsilon \end{aligned} \quad (6)$$

The test sample for model (6) includes audit firm years from fiscal years 2007 through 2010⁴⁶ and is restricted to auditors that have at least one client receiving media coverage.

⁴⁶ Requiring client *visibility* to be assessed two years prior to the test year eliminates two years of test observations compared to tests where *visibility* is assessed in the fiscal year of the audit. The timing of the *visibility* quintile classification was chosen so that it would precede the year of the restatement announcements and not be influenced by increased media coverage associated with restatements.

4. DATA SOURCES

Financial variable data are collected from Compustat, returns data are obtained from CRSP, audit fee, audit opinion, audit turnover, and restatement data are gathered from Audit Analytics, and AAER release date data is available on the SEC website.

The data used for measurement of business press attributes is obtained from RavenPack, a publicly available data source. RavenPack provides real-time news analytics on a broad number of textual business news sources primarily including newswires,⁴⁷ but also including newspapers such as *The Wall Street Journal*, major regional newspapers, and reputable blogs and other internet sources for financial news. The RavenPack data cover the years 2005 through May of 2011.

When a news story becomes available on these sources, RavenPack instantly analyzes the text and identifies all companies named. A separate observation is maintained for each company in the story. Each observation includes a relevance score based on the prominence of the company in the story. Relevance scores range from 0 to 100 with 100 being the most relevant.⁴⁸ Knowing how relevant a company is in a story aids in assessing how relevant other news analytics may be to that company. I limit my sample to observations with scores greater than or equal to 90. According to RavenPack this restriction retains only companies appearing in headlines and companies with

⁴⁷ RavenPack analyzes news stories from many different news sources, but their representatives indicate that the largest source of the data is newswires. The complete list of publishers and sources used by RavenPack is proprietary, but they do indicate that they include Dow Jones Newswires, PRNewswire, Businesswire, Globe Newswire, Marketwire, AP, CNW, The Wall Street Journal, Midnight Trader, Barron's, and more.

⁴⁸ Companies with identifiable roles in one of approximately 280 news categories (e.g. the firm is the plaintiff in a patent infringement lawsuit) are assigned relevance scores of 100. News categories are listed in Appendix B and are separated into 22 news groups. An example of why a company would receive a relevance score of 0 would be a firm's press release on a newswire that includes tags of other companies' names at the bottom of the press release. The companies not mentioned in the body of the text, only appearing as tags at the end are identified by RavenPack's algorithm, but receive low relevance scores.

identifiable roles from the category list⁴⁹ which includes 5,610,986 news story observations as shown on line one of Table 1.⁵⁰

⁴⁹ RavenPack instructed that companies named in headlines generally receive scores of 90 or more.

⁵⁰ As advised by RavenPack, I also remove all observations pertaining to market imbalance reports.

Table 1: Sample Composition

This table presents information on the number of observations used in the test samples. Analytics for news stories appearing from January 2005 through May 2011 are collected from RavenPack, a publicly available news analytics provider. Separate observations are created for each firm named in a news story and assigned a relevance score from 0 to 100, with higher scores corresponding to greater prominence of the firm in the story. News stories are classified as one of five news types including journalist generated full articles, news flashes, hot news flashes, tabular material, and firm generated press releases. RavenPack observations are condensed to firm year measures of *visibility* which is the natural log of one plus the number of articles about a firm and *prevalent sentiment* which is the average sentiment of articles covering a firm. Data from CRSP, Compustat, and Audit Analytics are merged to the RavenPack data.

	N
Individual US story company observations with <i>RELEVANCE</i> score greater than or equal to 90	5,610,986
Press release observations removed	4,413,682
Unique firm years with ISIN	34,491
Firm year observations with CRSP data and Compustat data*	20,626
Observations with Audit Analytics data and not missing data for computation of variables for tests of audit fees	17,356
Observations with data for computation of variables for audit opinion tests	16,960
Observations with data for computation of variables for audit opinion tests for distressed firms	5,751
Observations with data for computation of variables for auditor turnover tests	15,143
Observations with data for computation of variables for accruals tests	13,468
Observations with sufficient data to calculate one year changes for audit fee tests	12,889
* For years without coverage, <i>visibility</i> and <i>prevalent sentiment</i> measures are set to indicate no coverage and a neutral press sentiment.	

Stories are classified as one of five news types including full articles, press releases, hot news flashes, news flashes, and tabular material. I eliminate press release observations which are generated by firms in order to focus on the relationship between external media coverage and audit effort. This leaves 4,413,682 relevant news story observations. The sample is further reduced as I require observations to have an ISIN for merging RavenPack data with the Compustat, CRSP, and Audit Analytics databases.

After merging and removing observations missing data for the calculation of control variables, 17,356 firm year observations remain for use in the main tests using audit fees for the dependent variable. Additional data requirements reduce this sample for each of the supporting tests as denoted in Table 1.⁵¹ Each of these firm year observations combines information from multiple RavenPack observations to create news coverage attribute variables as described earlier.⁵² For firms without any press coverage, I set *visibility* values to correspond to zero coverage, and for additional tests *prevalent sentiment* values (which are described below) to correspond with a neutral media tone. As shown in Table 2, most stories in my sample are full articles with 2,245,105 observations, with the second greatest news type being regular news flashes having 1,641,450 observations. Table 2 also details descriptive statistics for the variables for the main tests of audit effort which use audit fees as a proxy.

⁵¹ To satisfy requirements for calculating media coverage attributes, firms with annual reporting dates before December 2005 or after February 2011 are deleted.

⁵² There are approximately 3,000,000 news story observations represented in the final test sample.

Table 2: Sample Data Descriptive Statistics

This table presents descriptive statistics for the test samples and variables. Analytics for news stories appearing from January 2005 through May 2011 are collected from RavenPack, a publicly available news analytics provider. Separate observations are created for each firm named in a news story and assigned a relevance score from 0 to 100, with higher scores corresponding to greater prominence of the firm in the story. Observations with relevance scores greater than or equal to 90 are retained. News stories are classified as one of five news types including journalist generated full articles, news flashes, and hot news flashes, tabular material, and firm generated press releases which are excluded. Each story is assigned five sentiment scores which are each calculated using dictionaries designed to capture sentiment of textual language more accurately in business settings. A composite of these scores (*CSS*) is also created for each score.

Panel A: News data composition

Panel A displays descriptive statistics for a linear transformation of *CSS* are provided by news type. Positive values of *CSS* denote a positive sentiment, negative values denote a negative sentiment, and zero indicates a neutral sentiment.

News Type	N	Mean CSS	Median CSS	Std Dev CSS
Full Article	2,245,105	-0.0036	0.0000	0.0473
Hot News Flash	62,865	-0.0192	0.0000	0.0853
News Flash	1,641,450	-0.0106	0.0000	0.0751
Tabular Material	464,262	-0.0049	0.0000	0.0215
Total	4,413,682			

Panel B: Test variables descriptive statistics

Panel B provides descriptive statistics for the test variables. Fee data are from Audit Analytics, news data are from RavenPack, financial data are from Compustat, and returns data are from CRSP. Refer to the Appendix A for variable definitions.

Variable	N	Mean	25th Pctl	Median	75th Pctl	Std Dev
<i>Acq</i>	17,356	0.484	0.000	0.000	1.000	0.500
<i>Big4</i>	17,356	0.694	0.000	1.000	1.000	0.461
<i>CRet_EQ</i>	17,356	1.020	0.832	0.977	1.141	0.340
<i>CRet_FY</i>	17,356	1.003	0.720	0.933	1.179	0.480
<i>Debt</i>	17,356	0.233	0.023	0.165	0.344	0.256
<i>Inventory</i>	17,356	0.092	0.000	0.025	0.141	0.132
<i>LitRisk</i>	17,356	0.273	0.000	0.000	1.000	0.446
<i>LnAssets</i>	17,356	6.407	4.973	6.443	7.777	2.064
<i>LnBadNewsVis</i>	17,356	2.180	1.609	2.197	2.773	0.996
<i>LnFee</i>	17,356	13.603	12.729	13.605	14.422	1.266
<i>LnGoodNewsVis</i>	17,356	2.499	1.946	2.485	3.045	0.965
<i>LnMVE</i>	17,356	6.058	4.625	6.016	7.407	1.999
<i>LnNeutralNewsVis</i>	17,356	3.136	2.485	3.135	3.738	0.907
<i>LnVis_EarlyQuarters</i>	17,356	3.826	3.258	3.807	4.317	0.841
<i>LnVis_FiscalYear</i>	17,356	4.516	3.970	4.489	4.970	0.803
<i>Loss</i>	17,356	0.417	0.000	0.000	1.000	0.493
<i>ModOpinion</i>	17,356	0.487	0.000	0.000	1.000	0.500
<i>PrevSent_EQ</i>	17,356	-0.010	-0.019	-0.005	0.004	0.022
<i>PrevSent_FY</i>	17,356	-0.010	-0.018	-0.006	0.002	0.019
<i>Receivables</i>	17,356	0.234	0.061	0.142	0.282	0.254
<i>RetVol_EQ</i>	17,356	-3.686	-4.097	-3.716	-3.311	0.580
<i>RetVol_FY</i>	17,356	-3.607	-4.010	-3.632	-3.231	0.562
<i>ROA</i>	17,356	0.029	0.007	0.053	0.117	0.205
<i>TradeVol_EQ</i>	17,356	-0.124	-0.592	-0.212	0.277	0.604
<i>TradeVol_FY</i>	17,356	-0.030	-0.485	-0.135	0.357	0.584

Five measures of sentiment are also assigned to each observation by RavenPack. Each of these five analytics indicates whether the tone of the story is negative, positive, or neutral.⁵³ They are described in greater detail in Appendix C. A sixth measure of tone is created using a proprietary combination of the first five scores. I use this composite of the other sentiment scores, the Composite Sentiment Score (*CSS*), as my proxy for the tone of each story in additional tests. I perform a linear transformation of *CSS* so it can be used as a measure of tone which ranges from -0.5 to 0.5 in value. Negative values indicate the story has a negative tone, *CSS* values of zero are considered neutral, and positive values are considered positive in tone. The mean *CSS* values for all news types are slightly negative and the median values are all zero corresponding with neutral tones for the median news story. A large portion of the scores are zero which supports the notion of an objective presentation of business press stories.

The *CSS* analytic is used to develop three new measures of *visibility* which partition news stories into negative, neutral, and positive tone subgroups. *BadNewsVis* is the natural log of one plus the number of stories published about a firm during the first two quarters of the fiscal year that had a negative *CSS* value. *GoodNewsVis* and *NeutralNewsVis* are calculated similarly. I also calculate measures corresponding to the full fiscal year for untabulated tests. These measures are implemented as additional analysis of the effect of media attributes other than “tone-deaf” *visibility*.⁵⁴

In additional tests *CSS* is also used to create two testable measures of a client’s coverage sentiment. *PrevSent_EarlyQuarters* is the mean *CSS* value for a firm’s coverage

⁵³ The five measures utilize separate dictionaries developed to assess tone in different business contexts. Recent research has highlighted that the use of general language dictionaries fails to account for the specialized meanings words can take in a business context (Loughran and McDonald, 2011).

⁵⁴ As stated above, I expect each of these measures to be positively associated with the effort proxy.

(the stories counted in *visibility* measures) for the first two quarters of the fiscal year, and a measure for the full fiscal year is also calculated. This measure is intended to capture the average tone of a firm's coverage.

Since I rely on *CSS* both to separate negative, positive, and neutral news stories for use in calculating tone conditional *visibility* scores as well as to assess the *prevalent sentiment* of a firm's media coverage, it is important to validate *CSS* as a measure of news story tone. I find that *CSS* is significantly positively associated with two day market adjusted returns calculated over the day a news story is published and the next day. Results from monthly panel regressions of *Ret*, which is the two day market adjusted return, on *CSS* are reported in Table 3.⁵⁵

⁵⁵ Results for one day market adjusted returns based on the day of the news story publication yield similar results.

Table 3: Regression of Two Day Market-adjusted Returns on *CSS*

This table presents the results from monthly panel regressions of two day market-adjusted returns (*Ret*) on the variable *CSS*. The test variable *CSS* is a linear transformation of a story's Composite Sentiment Score – a news story analytic collected from RavenPack. The linear transformation is described in Appendix A and the Composite Sentiment Score is described in Appendix C. *CSS* is a measure of a news story's sentiment based on counts of positive and negative words within each story's text. Positive values of *CSS* correspond to positive sentiment, negative values correspond to negative sentiment and values of zero indicate neutral sentiment. The day that a news story is released and the following day are used to calculate returns which are adjusted by the value weighted market return. The sample observations include the following news story types: full articles, news flashes, hot news flashes and tabular material. Observations have relevance scores of 90 or greater for years 2005 through 2010. News story types and the relevance analytic are described in Appendix C. There are a total of 3,236,258 news story observations used in the regression. Returns data are gathered from CRSP.

	Pred. sign	Coeff.	t-stat
Intercept		0.002	[3.9]
<i>CSS</i>	+	0.105	[30.8]
N	3,635,218		

Pearson correlation coefficients are presented in Table 4. The main dependent variable, *Ln_Fee* is significantly correlated with all of the test variables.⁵⁶ *Visibility* measures are positively correlated with measures of *prevalent sentiment*. This is consistent with the observation in Green, Hand, and Penn (2012) that there are more good news and neutral articles than bad news articles in the business press.

⁵⁶ Although the correlations are high, the variables are not perfectly collinear. In robustness tests I run regressions with collinearity diagnostics. Variance inflation factor values do not suggest that multicollinearity is problematic. The condition index value also does not raise concerns about multicollinearity. Some examples of firms with substantial size/*visibility* disparities are provided in Appendix F.

Table 4: Pearson Correlation Coefficients

This table provides correlations for the test variables. Fee data are from Audit Analytics, news data are from RavenPack, financial data are from Compustat, and returns data are from CRSP. Refer to the Appendix A for variable definitions.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)
(1) <i>Acq</i>	-																								
(2) <i>Big4</i>	0.24	-																							
(3) <i>CRet_EQ</i>	-0.01	0.03	-																						
(4) <i>CRet_FY</i>	0.00	0.07	0.69	-																					
(5) <i>Debt</i>	0.08	0.11	0.03	0.01	-																				
(6) <i>Inventory</i>	0.05	-0.01	0.06	0.04	-0.02	-																			
(7) <i>LitRisk</i>	0.04	0.05	0.02	0.01	-0.18	0.06	-																		
(8) <i>LnAssets</i>	0.25	0.45	0.00	0.05	0.26	-0.12	-0.23	-																	
(9) <i>LnBadNewsVis</i>	0.23	0.38	0.03	0.04	0.09	0.03	0.16	0.45	-																
(10) <i>LnFee</i>	0.42	0.60	0.03	0.06	0.18	0.03	-0.01	0.73	0.59	-															
(11) <i>LnGoodNewsVis</i>	0.28	0.36	0.04	0.05	0.13	0.01	0.10	0.55	0.59	0.63	-														
(12) <i>LnMVE</i>	0.29	0.55	0.12	0.22	0.14	-0.06	-0.04	0.82	0.55	0.79	0.64	-													
(13) <i>LnNeutralNewsVis</i>	0.29	0.42	0.06	0.08	0.13	0.02	0.09	0.57	0.70	0.68	0.68	0.68	-												
(14) <i>LnVis_EarlyQuarters</i>	0.30	0.43	0.03	0.04	0.13	0.02	0.11	0.58	0.78	0.70	0.81	0.69	0.91	-											
(15) <i>LnVis_FiscalYear</i>	0.30	0.44	0.05	0.07	0.14	0.02	0.12	0.60	0.81	0.72	0.84	0.72	0.94	0.96	-										
(16) <i>Loss</i>	-0.04	-0.11	-0.04	-0.08	0.02	-0.03	0.19	-0.36	0.06	-0.14	-0.13	-0.39	-0.15	-0.11	-0.11	-									
(17) <i>ModOpinion</i>	0.16	0.23	0.00	-0.01	0.04	0.05	0.09	0.11	0.19	0.31	0.19	0.22	0.20	0.20	0.22	0.01	-								
(18) <i>PrevSent_EQ</i>	0.08	0.12	0.05	0.03	0.03	0.00	-0.08	0.24	-0.15	0.15	0.33	0.31	0.20	0.19	0.19	-0.45	0.05	-							
(19) <i>PrevSent_FY</i>	0.08	0.13	0.10	0.12	0.04	0.00	-0.09	0.27	-0.19	0.17	0.38	0.36	0.23	0.20	0.21	-0.51	0.04	0.85	-						
(20) <i>Receivables</i>	-0.06	-0.29	-0.05	-0.03	-0.06	-0.10	-0.25	0.10	-0.28	-0.25	-0.18	-0.18	-0.24	-0.25	-0.26	-0.20	-0.22	0.08	0.09	-					
(21) <i>RetVol_EQ</i>	-0.12	-0.27	0.07	0.08	-0.10	0.08	0.18	-0.53	-0.12	-0.36	-0.26	-0.56	-0.24	-0.23	-0.25	0.48	-0.09	-0.36	-0.39	-0.08	-				
(22) <i>RetVol_FY</i>	-0.13	-0.28	0.01	-0.01	-0.07	0.07	0.16	-0.53	-0.12	-0.36	-0.25	-0.61	-0.25	-0.24	-0.25	0.51	-0.07	-0.36	-0.41	-0.06	0.91	-			
(23) <i>ROA</i>	0.15	0.15	0.10	0.14	0.05	0.14	-0.22	0.35	0.03	0.25	0.15	0.36	0.16	0.14	0.15	-0.49	0.02	0.27	0.34	0.11	-0.34	-0.37	-		
(24) <i>TradeVol_EQ</i>	-0.24	-0.46	-0.03	-0.05	-0.10	0.00	0.03	-0.58	-0.42	-0.61	-0.44	-0.69	-0.50	-0.50	-0.52	0.24	-0.20	-0.18	-0.20	0.24	0.40	0.40	-0.25	-	
(25) <i>TradeVol_FY</i>	-0.24	-0.48	-0.06	-0.09	-0.10	-0.01	0.03	-0.60	-0.42	-0.63	-0.45	-0.72	-0.51	-0.52	-0.53	0.26	-0.22	-0.21	-0.23	0.23	0.41	0.44	-0.28	0.90	-

Those significant at the less than 10% level appear in bold.

5. RESULTS

I find support for H1 both when measuring *visibility* over the first half of the fiscal year and when permitting it to cover the entire fiscal year. The results of these panel regressions are shown in Table 5 with the results for tests with the coverage variables measured over the first two fiscal quarters displayed in Panel A and those for the tests with fiscal year measured coverage variables presented in Panel B. The significant positive coefficients for measures of *visibility* are consistent with greater media coverage being associated with greater auditor effort as proxied for by audit fees. As with prior studies, the coefficient on *LnAssets* is positive and highly significant, and the adjusted R^2 is high at about 0.83 in both specifications.

In order to assess the economic significance of the association between *visibility* and auditor effort, I also run a modified version of these regressions. The design is the same as my regression reported in Table 5 Panel A except I take the annual decile rank of the *visibility* measure (ranging from zero for the lowest *visibility* to nine) divided by nine. This creates a *visibility* measure which ranges in value from zero to one with zero corresponding to the lowest level of *visibility*. The coefficient produced in this regression is then interpretable as a difference between the dependent variable value for the lowest and highest *visibility* firms.⁵⁷ In untabulated results, I find that the *visibility* coefficient is still positive and significant and has a value indicating an increase in the dependent variable effort proxy of approximately 18.7% for the highest *visibility* firms when compared to the lowest *visibility* firms. This difference does appear to be economically significant.

⁵⁷ I thank Mark Trombley for recommending this modification.

Table 5: Panel Regressions of Audit Effort on Media *Visibility*

This table provides results from panel regressions of the effort proxy *LnFee* on media coverage variables and other control variables. Year and 2-digit SIC industry fixed effects are included. Standard errors are clustered by firm. Fee data are from Audit Analytics, news data are from RavenPack, financial data are from Compustat, and returns data are from CRSP. *LnFee* is the natural log of audit fees. *LnVis_FiscalYear* is the natural log of one plus the number stories a firm appears in during the fiscal year. *Big4* is an indicator variable equal to one for firms audits performed by a Big 4 auditor and zero otherwise. *LnAssets* is the natural log of a firm's total assets. *LnMVE* is the natural log of the market value of equity. *Inventory* is total inventory divided by lagged total assets. *Receivables* is total receivables divided by lagged total assets. *Debt* is equal to the sum of total debt in current liabilities and total long-term debt divided by lagged total assets. *ROA* is operating income after depreciation divided by lagged total assets. *Loss* is an indicator variable equal to one if income before extraordinary items was negative in the current or either of the two previous years. *ModOpinion* is an indicator variable equal to one if the firm receives a modified audit opinion which is defined as anything other than a standard unqualified audit opinion coded as one in Compustat. *LitRisk* is an indicator variable equal to one if the firm belongs to a high litigation risk industry following Francis, Philbrick, and Schipper (1994). *Acq* is an indicator variable set equal to one if either acquisitions or restructuring costs after-tax are nonzero amounts. *CRet_FY* is equal to a firm's cumulative market adjusted return for the fiscal year. *RetVol_FY* is the natural log of the standard deviation of a firm's daily market adjusted returns for the fiscal year. *TradeVol_FY* is the standard deviation of a firm's daily trading volume for the fiscal year where trading volume is calculated as the day's trading volume scaled by the number of shares outstanding. *LnVis_EQ*, *CRet_EQ*, and *TradeVol_EQ* are calculated similarly to their corresponding fiscal year measures but measured over the first two fiscal quarters.

Panel A: Panel regression with the natural log of audit fees as the dependent variable and media *visibility* measured over the first two quarters of the fiscal year as the independent variable of interest

	Pred. Sign	Coeff.	t-stat
Intercept		10.005	[46.6]
<i>LnVis_EarlyQuarters</i>	+	0.156	[12.2]
<i>Big4</i>	+	0.349	[18.0]
<i>LnAssets</i>	+	0.409	[36.3]
<i>LnMVE</i>	+	0.052	[5.3]
<i>Inventory</i>	+	0.277	[3.5]
<i>Receivables</i>	+	0.163	[2.8]
<i>Debt</i>	+	-0.107	[-3.4]
<i>ROA</i>	-	-0.144	[-3.4]
<i>Loss</i>	+	0.202	[13.1]
<i>ModOpinion</i>	+	0.149	[12.8]
<i>LitRisk</i>	+	-0.057	[-1.8]
<i>Acq</i>	+	0.213	[15.9]
<i>Cret_EQ</i>	-	-0.037	[-3.0]
<i>RetVol_EQ</i>	+	0.064	[4.0]
<i>TradeVol_EQ</i>	+	-0.037	[-2.8]
Year FE		Yes	
Industry FE		Yes	
Firm Clustering		Yes	
N		17,356	
Adjusted R2		0.8341	

t-statistics appear in bold if significant at least at the 0.10 level for one-tailed tests.

Panel B: Panel regression with the natural log of audit fees as the dependent variable and media *visibility* measured over the fiscal year as the independent variable of interest

	Pred. Sign	Coeff.	t-stat
Intercept		10.035	[46.7]
<i>LnVis_FiscalYear</i>	+	0.168	[11.1]
<i>Big4</i>	+	0.346	[17.9]
<i>LnAssets</i>	+	0.395	[34.3]
<i>LnMVE</i>	+	0.070	[6.0]
<i>Inventory</i>	+	0.286	[3.6]
<i>Receivables</i>	+	0.164	[2.8]
<i>Debt</i>	+	-0.100	[-2.9]
<i>ROA</i>	-	-0.121	[-2.9]
<i>Loss</i>	+	0.194	[12.5]
<i>ModOpinion</i>	+	0.145	[12.6]
<i>LitRisk</i>	+	-0.066	[-2.1]
<i>Acq</i>	+	0.212	[15.8]
<i>Cret_FY</i>	-	-0.078	[-8.1]
<i>RetVol_FY</i>	+	0.107	[5.8]
<i>TradeVol_FY</i>	+	-0.045	[-3.2]
Year FE		Yes	
Industry FE		Yes	
Firm Clustering		Yes	
N		17,356	
Adjusted R2		0.8354	

t-statistics appear in bold if significant at least at the 0.10 level for one-tailed tests.

To further address the concern that bad news events are the actual cause of the primary test results I investigate the impact of tone on auditor effort. Three new measures of *visibility* are employed: *BadNewsVis*, *GoodNewsVis* and *NeutralNews_Vis* replace *LnVis_EarlyQuarters* in model (1). Each is predicted to be positive, but by measuring them separately auditor effort reactions to *visibility* are permitted to vary with the tone of coverage. Results reported in Table 6 show that, as predicted, each tone-sensitive *visibility* variable has a significant and positive coefficient.

Table 6: Panel Regression of Audit Effort on Tone Conditional Measures of Media Visibility

This table provides results from panel regressions of the audit effort proxy *LnFee* on media coverage variables and other control variables. Year and 2-digit SIC industry fixed effects are included. Standard errors are clustered by firm. Fee data are from Audit Analytics, news data are from RavenPack, financial data are from Compustat, and returns data are from CRSP. *LnFee* is the natural log of audit fees. *LnNeutralNewsVis* is the natural log or one plus the number of stories a firm appears in during the first two quarters of the fiscal year which have a neutral *CSS* value. *LnBadNewsVis* and *LnGoodNewsVis* are calculated similarly but for stories where *CSS* value correspond to negatively toned articles and positively toned stories respectively. Other variables are defined in Table 5 as well as in the Appendix A.

Panel regression with the natural log of audit fees as the dependent variable and media tone conditional *visibility* variables measured over the first two quarters of the fiscal year

	Pred. Sign	Coeff.	t-stat
Intercept		10.169	[48.4]
<i>LnNeutralNewsVis</i>	+	0.071	[6.8]
<i>LnBadNewsVis</i>	+	0.042	[5.6]
<i>LnGoodNewsVis</i>	+	0.054	[6.3]
<i>Big4</i>	+	0.350	[18.0]
<i>LnAssets</i>	+	0.407	[35.9]
<i>LnMVE</i>	+	0.049	[5.0]
<i>Inventory</i>	+	0.278	[3.5]
<i>Receivables</i>	+	0.166	[2.8]
<i>Debt</i>	+	-0.107	[-3.1]
<i>ROA</i>	-	-0.140	[-3.4]
<i>Loss</i>	+	0.199	[12.9]
<i>ModOpinion</i>	+	0.148	[12.8]
<i>LitRisk</i>	+	-0.060	[-1.9]
<i>Acq</i>	+	0.213	[15.6]
<i>Cret_EQ</i>	-	-0.038	[-3.1]
<i>RetVol_EQ</i>	+	0.057	[3.6]
<i>TradeVol_EQ</i>	+	-0.043	[-3.1]
Year FE		Yes	
Industry FE		Yes	
Firm Clustering		Yes	
N		17,356	
Adjusted R2		0.8343	

t-statistics appear in bold if significant at least at the 0.10 level for one-tailed test.

Further evidence supporting the other auditor response hypotheses is provided through tests of the effect of *visibility* on audit opinion modifications, accrual values, and auditor turnover. As shown in the first two columns of Table 7, *visibility* is positively related to the probability of the auditor issuing a modified audit opinion when tested in a sample of all firm year observations (Column 1) as well as when the sample is restricted to distressed firms (Column 2). The probability of issuing a going concern opinion in the distressed firms subsample is also positively associated with *visibility*. Unlike with the tests of audit fees, an audit opinion is not final until the end of the engagement, so it is not necessary to evaluate the tests using *LnVis_EarlyQuarters* as a measure of client firm media *visibility*.

Table 7: Logistic Regressions of Audit Opinion Variables on Media *Visibility*

This table provides results from logistic regressions of audit opinion variables on media coverage variables and other control variables. Year fixed effects are included. Audit report data are from Audit Analytics, news data are from RavenPack, financial data are from Compustat, and returns data are from CRSP. The modified audit opinion dependent variable *ModOpinion* is an indicator variable equal to one if the firm receives a modified audit opinion which is defined as anything other than a standard unqualified audit opinion coded as one in Compustat. The going concern opinion dependent variable *GConcern* is an indicator variable equal to one if the firm receives a going concern opinion in the audit report. *LnVis_FiscalYear* is the natural log of one plus the number stories a firm appears in during the fiscal year. *Sales* is the natural log of total revenues. *ChgDebt* is the change in long term debt scaled by lagged total assets. *Prnocf* is an indicator variable equal to one if the firm had negative operating cash flows in the previous year. *Delay* is the number of days between the fiscal year end and the audit report signature date. *NewDebt* is an indicator variable equal to one if the company issues new debt in the following year. Other variables are defined in Table 5 as well as in the Appendix A.

Independent variables	Dependent variable Pred. Sign	= Modified audit opinion		Going concern opinion
		Coeff. (Pr > χ^2)	Coeff. (Pr > χ^2)	Coeff. (Pr > χ^2)
Intercept		-4.430 (<0.0001)	-4.308 (<0.0001)	-2.022 (0.0377)
<i>LnVis_FiscalYear</i>	+	0.238 (<0.0001)	0.154 (0.0061)	0.667 (<0.0001)
<i>Sales</i>	-	0.136 (<0.0001)	0.043 (0.0397)	0.017 (.3690)
<i>LnMVE</i>	?	0.054 (0.0010)	0.114 (0.0011)	-0.767 (<0.0001)
<i>ROA</i>	-	-0.142 (0.1286)	-0.011 (0.4760)	-3.017 (<0.0001)
<i>Debt</i>	+	-0.045 (0.2900)	0.310 (0.0057)	1.537 (<0.0001)
<i>ChgDebt</i>	?	-0.065 (0.3455)	-0.673 (0.0036)	-2.889 (<0.0001)
<i>Loss</i>	+	0.327 (<0.0001)	0.484 (<0.0001)	0.416 (0.2454)
<i>Prnocf</i>	+	0.246 (<0.0001)	0.168 (0.0089)	0.444 (0.0055)
<i>Big4</i>	+	0.700 (<0.0001)	0.787 (<0.0001)	0.469 (0.0009)
<i>Delay</i>	+	0.015 (<0.0001)	0.020 (<0.0001)	0.028 (<0.0001)
<i>NewDebt</i>	-	-0.200 (<0.0001)	-0.150 (0.0109)	0.364 (0.0039)
<i>Cret_FY</i>	-	-0.028 (0.2391)	-0.151 (0.0073)	-0.373 (0.0052)
<i>RetVol_FY</i>	?	0.235 (<0.0001)	0.233 (0.0070)	1.306 (<0.0001)
<i>TradeVol_FY</i>	?	-0.450 (0.0001)	-0.363 (<0.0001)	-0.793 (<0.0001)
Year FE		Yes	Yes	Yes
N		16,960	5,751	5,751
Percent concordant		75.8	74.9	90.2
Pseudo R2		0.263	0.240	0.390

χ^2 probabilities reported for one-tailed tests.

The results from the test of signed accruals are displayed in Table 8. The coefficient on *LnVis_FiscalYear* is -0.008 and is statistically significant with a t-statistic of -4.4. This result is consistent with auditors influencing accruals downward. This is the expected relationship for auditors responding to the uncertainty of accruals and the increased risk pertaining to client *visibility* by asymmetrically requiring downward adjustments of accruals in audited financial statements. This result serves as another piece of evidence that auditors respond to media coverage of a client firm.

Table 8: Panel Regression of Accruals Variables on Media *Visibility*

This table provides results from panel regressions of accruals variables on media coverage variables and other control variables. Year and 2-digit SIC industry fixed effects are included. Standard errors are clustered by firm. News data are from RavenPack, financial data are from Compustat, and returns data are from CRSP. The dependent variable *Acc* is the signed accruals scaled by lagged total assets. *LnVis_FiscalYear* is the natural log of one plus the number stories a firm appears in during the fiscal year. *PPE* is the lagged value of net property, plant and equipment scaled by lagged total assets. *AssetGrowth* is total assets less the lagged value of total assets scaled by lagged total assets. *DSales* is total revenue less lagged total revenue scaled by lagged total assets. *CFO* is net operating cash flow scaled by lagged total assets. *MB* is the market value of equity divided by the book value of common equity. Other variables are defined in Table 5 as well as in the Appendix A.

Independent variables	Coeff. (t-stat)
Intercept	-0.135 (-9.89)
<i>LnVis_FiscalYear</i>	-0.008 (-4.37)
<i>LnAssets</i>	0.007 (4.01)
<i>LnMVE</i>	-0.006 (-3.34)
<i>PPE</i>	0.004 (1.00)
<i>AssetGrowth</i>	0.069 (11.94)
<i>DSales</i>	-0.024 (-5.11)
<i>ROA</i>	0.717 (38.64)
<i>CFO</i>	-0.812 (-42.27)
<i>MB</i>	0.000 (-1.26)
<i>Debt</i>	-0.068 (-10.97)
<i>Loss</i>	-0.019 (-7.56)
<i>Cret_FY</i>	0.022 (10.06)
<i>RetVol_FY</i>	-0.034 (-10.75)
<i>TradeVol_FY</i>	0.012 (4.56)
Year FE	Yes
Industry FE	Yes
Firm Clustering	Yes
N	13,468
Adjusted R2	0.4846

As reported in Table 9, results for tests of turnover confirm predictions that auditor turnover is positively related to levels of media coverage. Controlling for several client characteristics (size, market value, and performance) and auditor attributes (size and tenure), *LnVis_FiscalYear* has a significantly positive coefficient. Interpreted with the previous tests of audit fees, opinion modifications, and reported accruals, it appears that auditors do respond to media coverage of audit clients by increasing audit effort and altering other audit decisions. The probability of an auditor client relationship ending also increases significantly.

Table 9: Logistic Regression of Auditor Turnover on Media *Visibility*

This table provides results from logistic regressions of *Turnover* on media coverage variables and other control variables. Year fixed effects are included. Audit tenure data are from Audit Analytics, news data are from RavenPack, financial data are from Compustat, and returns data are from CRSP. The dependent variable *Turnover* is an indicator variable equal to one if there is an auditor change in the subsequent year. *LnVis_FiscalYear* is the natural log of one plus the number stories a firm appears in during the fiscal year. *Growth* is total sales less lagged sales scaled by lagged sales. *ShortTenure* is an indicator variable equal to one if the auditor has worked less than three years with the client. *LongTenure* is an indicator variable equal to one if the auditor has worked more than eight years with the client. Other variables are defined in Table 5 as well as in the Appendix A.

Independent variables	Pred. Sign	Coeff. (Pr > χ^2)
Intercept		-8.817 (<0.0001)
<i>LnVis_FiscalYear</i>	+	0.166 (0.0191)
<i>LnAssets</i>	-	-0.030 (0.2230)
<i>LnMVE</i>	-	-0.235 (<0.0001)
<i>Sales</i>	-	0.005 (0.4508)
<i>ROA</i>	-	-0.299 (0.0975)
<i>Growth</i>	-	-0.222 (0.0099)
<i>Debt</i>	+	-0.151 (0.1636)
<i>Big4</i>	?	0.749 (<0.0001)
<i>ShortTenure</i>	+	6.801 (<0.0001)
<i>LongTenure</i>	+	2.0852 (0.0230)
<i>Cret_FY</i>	-	0.112 (0.0915)
<i>RetVol_FY</i>	?	-0.143 (0.1158)
<i>TradeVol_FY</i>	?	0.038 (0.3440)
Year FE		Yes
N		15,143
Percent concordant		87.2
Pseudo R2		0.3402

χ^2 probabilities reported for one-tailed tests.

5.1 Additional Analyses Results

Table 10 shows that there is also a significant relationship in the predicted direction for *PrevSent_FiscalYear* when it is added to model (1). This evidence supports the hypothesis that auditor effort is greater for firms with more negatively toned coverage on average. The sign for the *visibility* measure is still positive and significant.

Table 10: Panel Regression of Audit Effort on Media *Visibility* and *Prevalent Sentiment*

This table provides results from panel regressions of audit effort proxy *LnFee* on media coverage variables and other control variables. Year and 2-digit SIC industry fixed effects are included. Standard errors are clustered by firm. Fee data are from Audit Analytics, news data are from RavenPack, financial data are from Compustat, and returns data are from CRSP. *LnFee* is the natural log of audit fees. *LnVis_FiscalYear* is the natural log of one plus the number stories a firm appears in during the fiscal year. *PrevSent_FY* is the mean Composite Sentiment Score (CSS) value of the stories included in *LnVis_FiscalYear*. *LnNeutralNewsVis* is the natural log or one plus the number of stories a firm appears in during the fiscal year which have a neutral CSS value. *LnBadNewsVis* and *LnGoodNewsVis* are calculated similarly but for stories where CSS value correspond to negatively toned articles and positively toned stories respectively. Other variables are defined in Table 5 as well as in the Appendix A.

Panel regression with the natural log of audit fees as the dependent variable and media *visibility* and *prevalent sentiment* measured over the first two quarters of the fiscal year

	Pred. Sign	Coeff.	t-stat
Intercept		9.977	[46.2]
<i>LnVis_EarlyQuarters</i>	+	0.159	[12.3]
<i>PrevSent_EarlyQuarters</i>	-	-0.725	[-3.0]
<i>Big4</i>	+	0.349	[18.0]
<i>LnAssets</i>	+	0.408	[36.0]
<i>LnMVE</i>	+	0.054	[5.4]
<i>Inventory</i>	+	0.278	[3.5]
<i>Receivables</i>	+	0.165	[2.8]
<i>Debt</i>	+	-0.105	[-3.1]
<i>ROA</i>	-	-0.138	[-3.3]
<i>Loss</i>	+	0.191	[12.3]
<i>ModOpinion</i>	+	0.150	[12.9]
<i>LitRisk</i>	+	-0.057	[-1.9]
<i>Acq</i>	+	0.214	[16.0]
<i>Cret_EQ</i>	-	-0.035	[-2.8]
<i>RetVol_EQ</i>	+	0.061	[3.8]
<i>TradeVol_EQ</i>	+	-0.036	[-2.8]
Year FE		Yes	
Industry FE		Yes	
Firm Clustering		Yes	
N		17,356	
Adjusted R2		0.8342	

t-statistics appear in bold if significant at least at the 0.10 level for one-tailed test.

As a stricter test of the relationship between media *visibility* and auditor effort, I test changes in audit fees regressed on changes in fee determinant variables, I find the same support when using a changes design for the audit fee variable.⁵⁸ Results for tests of early quarters measures of *visibility* and fiscal year measures are presented in panels A and B of Table 11. The coefficient on the audit fee variable is positive and significant in each of these tables. In Panel C, results are reported for a changes design using tone-partitioned measures of *visibility*. The signs for changes in *NeutralNewsVis* and *GoodNewsVis* are both positive and significant. The coefficient for changes in *BadNewsVis* is negative but is not even marginally significant.

⁵⁸ Changes models can be problematic for tests of audit fees due to the step-function nature of the fee setting process. A one dollar change in the value of a firm's total assets, the major predictor of audit fees, does not have to have any impact on the audit fee. My variables of interest are conceptually appropriate for a changes design because small fluctuations in coverage can be indicative of a change in the anticipated auditor business risk as a company moves into or out of the public eye. I appreciate Robert Knechel's advice on this specification concern.

Table 11: Panel Regressions of Changes in Audit Effort on Changes in Media *Visibility*

This table provides results from panel regressions of D_LnFee on media coverage variables and other control variables. Year and 2-digit SIC industry fixed effects are included. Standard errors are clustered by firm. Fee data are from Audit Analytics, news data are from RavenPack, financial data are from Compustat, and returns data are from CRSP.

D_LnFee is the change in the natural log of audit fees from the previous year. $D_LnVis_FiscalYear$ is the change in the natural log of one plus the number stories a firm appears in during the fiscal year from the previous year.

$D_LnVis_EarlyQuarters_PrevSent_FY$ is the mean Composite Sentiment Score (CSS) value of the stories included in $LnVis_FiscalYear$. $LnNeutralNewsVis$ is the natural log or one plus the number of stories a firm appears in during the fiscal year which have a neutral CSS value. $LnBadNewsVis$ and $LnGoodNewsVis$ are calculated similarly but for stories where CSS values correspond to negatively toned articles and positively toned stories respectively. Other variables are defined in Table 5 as well as in the Appendix A.

Panel A: Panel regression with the change in the natural log of audit fees from the previous year as the dependent variable and media *visibility* measured over the first two quarters of the fiscal year

	Pred. Sign	Coeff.	t-stat
Intercept		-0.098	[-0.8]
$D_LnVis_EarlyQuarters$	+	0.027	[4.6]
D_Big4	+	0.240	[9.4]
$D_LnAssets$	+	0.263	[16.7]
D_LMVE	+	-0.016	[-2.7]
$D_Inventory$	+	0.128	[2.0]
$D_Receivables$	+	-0.049	[-1.3]
D_Debt	+	0.019	[1.0]
D_ROA	-	-0.107	[-3.2]
D_Loss	+	0.038	[5.2]
$D_ModOpinion$	+	-0.013	[-2.6]
D_Acq	+	0.027	[4.8]
D_Cret_EQ	-	-0.003	[-0.5]
D_RetVol_EQ	+	0.003	[0.4]
$D_TradeVol_EQ$	+	0.012	[2.0]
Year FE		Yes	
Industry FE		Yes	
Firm Clustering		Yes	
N		12,889	
Adjusted R2		0.1272	

t-statistics appear in bold if significant at least at the 0.10 level for one-tailed test.

Panel B: Panel regression with the change in the natural log of audit fees from the previous year as the dependent variable and media *visibility* measured over the fiscal year

	Pred. Sign	Coeff.	t-stat
Intercept		-0.094	[-0.8]
<i>D_LnVis_FiscalYear</i>	+	0.052	[6.7]
<i>D_Big4</i>	+	0.239	[9.3]
<i>D_LnAssets</i>	+	0.245	[14.8]
<i>D_LMVE</i>	+	0.002	[0.3]
<i>D_Inventory</i>	+	0.124	[1.9]
<i>D_Receivables</i>	+	-0.041	[-1.1]
<i>D_Debt</i>	+	0.025	[1.2]
<i>D_ROA</i>	-	-0.107	[-3.2]
<i>D_Loss</i>	+	0.040	[5.4]
<i>D_ModOpinion</i>	+	-0.013	[-2.6]
<i>D_Acq</i>	+	0.026	[4.5]
<i>D_Cret_FY</i>	-	-0.022	[-3.9]
<i>D_RetVol_FY</i>	+	0.008	[0.8]
<i>D_TradeVol_FY</i>	+	0.010	[1.6]
Year FE		Yes	
Industry FE		Yes	
Firm Clustering		Yes	
N		12,889	
Adjusted R2		0.1314	

t-statistics appear in bold if significant at least at the 0.10 level for one-tailed test.

Panel C: Panel regression with the change in the natural log of audit fees from the previous year as the dependent variable and changes in media tone conditional *visibility* measures estimated for the first two quarters of the fiscal years

	Pred. Sign	Coeff.	t-stat
Intercept		-0.100	[-0.8]
<i>D_LnNeutralNewsVis</i>	+	0.017	[3.4]
<i>D_LnBadNewsVis</i>	+	-0.003	[-1.0]
<i>D_LnGoodNewsVis</i>	+	0.015	[3.8]
<i>D_Big4</i>	+	0.240	[9.4]
<i>D_LnAssets</i>	+	0.262	[16.6]
<i>D_LMVE</i>	+	-0.016	[-2.7]
<i>D_Inventory</i>	+	0.127	[2.0]
<i>D_Receivables</i>	+	-0.048	[-1.3]
<i>D_Debt</i>	+	0.018	[0.9]
<i>D_ROA</i>	-	-0.110	[-3.3]
<i>D_Loss</i>	+	0.040	[5.4]
<i>D_ModOpinion</i>	+	-0.013	[-2.5]
<i>D_Acq</i>	+	0.027	[4.7]
<i>D_Cret_EQ</i>	-	-0.004	[-0.5]
<i>D_RetVol_EQ</i>	+	0.002	[0.3]
<i>D_TradeVol_EQ</i>	+	0.012	[2.0]
Year FE		Yes	
Industry FE		Yes	
Firm Clustering		Yes	
N		12,889	
Adjusted R2		0.1281	

t-statistics appear in bold if significant at least at the 0.10 level for one-tailed test.

5.2 Evidence of Auditor Costs of Client Media Coverage

To provide evidence of the costs of client media coverage to auditors, I test the association between a client's fiscal year *visibility* and the likelihood of the SEC releasing an AAER for the client in the following year. Results for this test are reported in Table 12 below. The coefficient for *LnVis_FiscalYear* is positive and significant with a coefficient of 0.005 and a t-statistic of 3.9. The adjusted R^2 for the model is low at 0.0084. I do also perform a robustness test of the relationship including all the additional auditor risk and newsworthiness variables from tests of model (1) and find the same qualitative results for the variable of interest. This evidence is consistent with media *visibility* having a potential negative effect on the auditor's reputation as they are associated with a client that is more likely to receive an AAER.

Table 12: Panel Regression of Client AAERs on Prior Media *Visibility*

This table provides results from a panel regression of *AAER* on a media coverage variable and other control variables. Year and 2-digit SIC industry fixed effects are included. Standard errors are clustered by firm. SEC Accounting and Auditing Enforcement Release (AAER) date data are available on the SEC website, news data are from RavenPack, and financial data are from Compustat. *AAER* is an indicator variable set equal to one if an AAER is released in year $t+1$. *LnVis_FiscalYear* is the natural log of one plus the number stories a firm appears in during the fiscal year. Other variables are defined in Table 5 as well as in Appendix A.

	Pred. Sign	Coeff.	t-stat
Intercept		0.003	[0.1]
<i>LnVis_FiscalYear</i>	+	0.005	[3.9]
<i>LnAssets</i>		0.003	[2.9]
<i>LnMVE</i>		-0.001	[-1.6]
<i>ROA</i>		0.000	[0.0]
<i>Debt</i>		-0.008	[-3.3]
<i>Big4</i>		-0.001	[-0.9]
<i>MB</i>		0.000	[0.0]
<i>Loss</i>		0.003	[2.2]
<i>AbsAcc</i>		0.002	[0.4]
Year FE		Yes	
Industry FE		Yes	
Firm Clustering		Yes	
N		15,212	
Adjusted R2		0.0084	

t-statistics appear in bold if significant at least at the 0.10 level for one-tailed tests where signs are predicted.

As a final cost test, I examine the relationship between high *visibility* client restatements and an auditor's total audit fee revenue collected from all clients in a fiscal year. The results reported in Table 13 below indicate a significant negative association between the variable of interest, *LnHighVisRestate*, and the dependent variable. This is consistent with the prediction that as the collective magnitude of restatements by an auditor's highly visible clients increases, the reputational and other costs experienced by the auditor actually lead to a decrease in the total fee revenue they collect annually. The coefficient for low *visibility* client restatements is insignificant. All other control variables are significant with coefficient signs in the predicted directions. The adjusted R^2 for the model is 0.9173. As a robustness test, I also include a variable capturing the magnitude of restatements by clients in the middle three *visibility* quintiles. The results for the variable of interest are qualitatively the same in this specification.

Table 13: Panel Regression of Total Auditor Fee Revenues on High *Visibility* Restatement Totals

This table provides results from a panel regression of *LnFeeRev* on the absolute value of an auditor's clients' individual net restatements for high *visibility* clients (*LnHighVisRestate*). Controls are included for low *visibility* client restatements (*LnLowVisRestate*) as well as for total auditor client *visibility* (*LnAuditorVis*), prior year total auditor fee revenues (*LnLagFeeRev*), and auditor size (*Big4*). Year fixed effects are included. Standard errors are clustered by audit firm. Fee data are from Audit Analytics and news data are from RavenPack. *LnFeeRev* is the natural log of the total audit fees received by an auditor from all clients in the fiscal year. *LnHighVisRestate* is the natural log of the total of the absolute value of net dollars restated by an auditor's client firms in the fiscal year *t*-1 where the client firm was in the top *visibility* quintile in fiscal year *t*-2. *LnLowVisRestate* is the natural log of the total of the absolute value of net dollars restated by an auditor's client firms in the fiscal year *t*-1 where the client firm was in the bottom *visibility* quintile in fiscal year *t*-2. *LnAuditorVis* is the natural log of the total number of stories published about an auditor's clients during the fiscal year. *LnLagFeeRev* is the prior year value of *LnFeeRev*. *Big4* is an indicator variable equal to one for firms audited by a Big 4 auditor and zero otherwise. The sample observations are limited to auditor firm years for auditors with clients receiving media coverage. Lag requirements restrict the sample to fiscal years 2007 through 2010.

	Pred. Sign	Coeff.	t-stat
Intercept		2.953	[7.2]
<i>LnHighVisRestate</i>	-	-0.007	[-2.6]
<i>LnLowVisRestate</i>		0.001	[0.2]
<i>LnAuditorVis</i>	+	0.344	[6.2]
<i>LnLagFeeRev</i>	+	0.672	[14.7]
<i>Big4</i>	+	0.358	[2.8]
Year FE		Yes	
Audit Firm Clustering		Yes	
N		736	
Adjusted R2		0.9173	

t-statistics appear in bold if significant at least at the 0.10 level for one-tailed tests where signs are predicted.

5.3 Robustness

The changes tests reported in Table 11 provide greater assurance that the relationship between *visibility* and auditor effort is robust. For additional evidence, I perform several alternate tests on the primary auditor decision investigated – auditor effort. I first estimate the model (1) regression by year and size decile.⁵⁹ I also estimate the model as a rank regression, and then as a quantile regression at the median. These tests are performed using *visibility* measures calculated both over the first two quarters and the full fiscal year. The sign for measures of *visibility* remains positive through all of these regressions and is statistically significant.

To investigate whether effort increases related to news coverage from the previous year are eventually priced I regress lagged *visibility* on current year audit fees and find a significant positive coefficient for *visibility*. This result also holds when combining lagged measures of *visibility* with an additional variable that captures *visibility* during the first quarter of the fiscal year.

As greater control for the underlying news events I also include the independent variable *Group* which is a vector of twenty-two indicator variables which take on a value of one if the firm had a news story released during the estimation year which corresponded to one of the twenty-two news story group classifications listed in Appendix B. This variable is not found in typical audit fee models, but again, I am trying to separate the effect of media coverage from the nature of the underlying news events. The findings are robust to inclusion of *Groups*.

As accruals are noted to be challenging to audit due to their uncertainty, I also

⁵⁹ Picconi and Reynolds (2009) find that this improves both the estimation and explanatory power of audit fee models that regress the natural log of audit fees on the natural log of assets and other independent variables.

include signed and absolute accruals separately and find that the results for the variable of interest are unaffected.

There is also potential for the non-audit fees received from a client to influence the level of auditor effort. Independence impairments are less of a concern since my sample begins after passage of the Sarbanes-Oxley Act, but I do perform tests which include the natural log of non-audit fees as an independent variable and find the same results for *visibility* measures. I also repeat the regressions using the natural log of total fees as the dependent variable and the results are qualitatively the same.

Another concern is the strong correlation between media coverage variables and a client firm's size. As noted in footnote 56, test diagnostics of multicollinearity do not indicate that there is a problem, and the variables, though highly correlated, are not perfectly collinear. Still, I attempt to better control for size by using alternate measures of firm size separately and then together. Using total assets, total revenues, and market capitalization each independently and then collectively does not qualitatively affect the results for *visibility* measures.

As mentioned above, I estimate the model by year and size decile. Additionally I increase the number of size partitions beyond ten and find that as the number is increased, measures of size lose significance while the *visibility* coefficients remain significant for as many as ninety-nine size groupings.⁶⁰

Several other specifications for the other auditor decisions have also been checked. Tests of audit opinions are robust to measurement with standard errors clustered

⁶⁰ The failure to find significance for size as an independent variable is consistent with the notion that audit fees are estimated as a step function. Consequently, when the number of size groups is increased, the significance of size diminishes even though it is a major determinant of audit fees. I thank Robert Knechel for this observation.

by firm. Coefficient signs and significance are also qualitatively the same when specified as linear probability regressions with standard errors clustered by firm and year fixed effects included. Turnover regressions have been run using total asset size quintile and decile dummy variables in place of *LnAssets*. Results are also robust to alternate specifications of the control for trading volume. Calculating trading volume as the number of days that volume exceeds the mean volume for the year, and calculating volume as the standard deviation of daily share turnover during the year do not affect the variable of interest.

Some final specifications include: removing financial industries from all tests and removing firms with absolute accruals greater than total assets; limiting the sample to a single observation per firm; expanding the news stories to include all articles with relevance values greater than or equal to 75 (indicating the firm was identified by the end of the first paragraph of a news story); and restricting news stories to include only articles with relevance values equal to 100 (indicating that the firm has an identified role in RavenPack's category index of events of interest).

6. CONCLUSION

This study analyzes the effect of the business press (a key component of the information environment) on the effort and effectiveness of the institutional monitoring (by auditors) of financial reporting. Using the number of articles covering a client firm (a proxy for *visibility*) I find that auditors charge higher audit fees for clients with higher *visibility* incremental to other determinants of audit fees. The auditor effort evidence using fees as a proxy is corroborated using other proxies for auditor decisions: audit report modifications, the level of accruals, and auditor turnover. Proving robust to several alternate specifications, the evidence supports the contention that there is a significant relationship between business press coverage and auditor decisions. My findings show that as the media performs increased monitoring of individual firms, external auditors do respond by working harder, increasing their propensity to modify audit opinions, negotiating lower reported accruals, and terminating the engagement with greater probability.

I test the impact of media tone and find that auditor effort increases in response to measures of positive, negative, and neutral *visibility*. Additionally, I find limited evidence that audit effort is negatively related to the average tone (*prevalent sentiment*) of coverage. Supporting the predictions that auditors respond to press coverage, I also document two potential costs auditors of high *visibility* clients may incur. I find that the probability of a client receiving an AAER is increasing in media *visibility*, and I find that the total audit fee revenue an auditor collects from all its clients is decreasing in the magnitude of restatements by high *visibility* client firms.

Collectively, the evidence suggests that the business press has an important effect

on auditor decisions. I conclude that the press not only participates in the external monitoring of financial reporting, but that through its influence on official external monitors the press actually also affects financial reporting.

These findings have implications for multiple parties. The observed associations between client *visibility* and decisions made by auditors is at odds with claims made by some practicing auditors that auditor business risk and media coverage of client firms does not influence the conduct of the audit.^{61,62} Consequently, client firms should be aware of the additional scrutiny and other effects that are associated with their own *visibility*. When making decisions to influence their prominence in the media, client firms should take into account the result that coverage of all tones is associated with increased levels of auditor effort. Policy makers and enforcement authorities should consider both how auditors respond to firm *visibility* and how media coverage is associated with their enforcement efforts. Similarly, the business press should recognize its potentially influential role in the external monitoring of financial reporting. Auditors can be informed regarding the reputational and monetary costs associated with the *visibility* of their clients. Finally, these results have implications for auditing standards and may need to be addressed by standard setters.

⁶¹ In interviews conducted with current and former auditors ranging in experience levels from new hires to partner level, opinions on the effect and extent of influence of client *visibility* on audit performance varied. The suggestion that *visibility* would have no noticeable effect on audit performance is not supported by my results. This view was shared by some practitioners including at the partner level.

⁶² See footnote 17 for a brief discussion of auditor business risk's influence on audit planning and performance.

6.1 Research Agenda

In a concurrent working paper, Green, Hand, and Penn (2012) find evidence that although there are a greater number of business press articles that are positive and neutral in tone relative to bad news articles, there does exist a bad news bias in the business press. We find bad news stories receive a greater number of follow up stories than good news stories. In addition to documenting the presence of bias, we provide evidence supporting two different causes and document how bias affects market reactions to the release of news stories.

In this study I seek to understand in general how auditors respond to increased coverage levels for firms. The Green, Hand, and Penn (2012) findings suggest that as firms receive greater press coverage, they are both subjected to more biased coverage and are greater scrutinized by investors. I expect auditors to increase their effort in anticipation of this increased scrutiny. Subsequent to this paper, I propose to further examine the relationship between media coverage and auditor effort by considering issues such as the effect of auditor expertise on the influence of media coverage, as well as the influence of media source on auditor responses.⁶³ Some other areas for future research include the internal control audit, and the possible contagion effects of industry level media measures on audit engagements.

⁶³ Joe (2002) suggests future research could consider the influence of redundant positive information on auditor decisions, how the source of the redundant information affects its influence on auditors, and how the type and quantity of press coverage affect the influence of redundant information.

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APPENDIX A – Variable Definitions

Variable	Definition
<i>AAER</i>	An indicator variable set equal to one if the SEC issues an Accounting and Auditing Enforcement Release in the subsequent year.
<i>AbsAcc</i>	The absolute value of total accruals scaled by lagged total assets.
<i>Acc</i>	The signed value of total accruals scaled by lagged total assets.
<i>Acq</i>	An indicator variable set equal to one if either acquisitions or restructuring costs after-tax are nonzero amounts.
<i>AssetGrowth</i>	Total assets less the lagged value of total assets scaled by lagged total assets.
<i>Big4</i>	An indicator variable set equal to one for firms audited by a Big 4 auditor and zero otherwise.
<i>CFO</i>	Net operating cash flow scaled by lagged total assets.
<i>ChgDebt</i>	The change in long-term debt divided by lagged total assets.
<i>CRet_EQ</i>	A firm's cumulative market adjusted return for the first two quarters of the fiscal year.
<i>CRet_FY</i>	A firm's cumulative market adjusted return for the fiscal year.
<i>CSS</i>	RavenPack's news article sentiment analytic <i>Composite Sentiment Score</i> transformed by subtracting fifty and dividing by one hundred.
<i>D_</i>	Prefix indicating that a variable is measured as a change (current year less previous year value).
<i>Debt</i>	The sum of total debt in current liabilities and total long-term debt divided by lagged total assets.
<i>Delay</i>	The number of days between the fiscal year end and the auditor's report signing date.
<i>DSales</i>	Total revenue less lagged total revenue scaled by lagged total assets.
<i>GConcern</i>	An indicator variable equal to one if the firm receives a going concern opinion in the audit report.
<i>Group</i>	A set of twenty-two indicator variables corresponding to the news story groups listed in Appendix B. The value of a group variable is set equal to one if a news story corresponding to the group is published during the year.
<i>Growth</i>	Total sales less lagged sales scaled by lagged sales.
<i>Ind</i>	Fixed effect indicating the observation firm's two-digit SIC.
<i>Inventory</i>	Total inventory divided by lagged total assets.

APPENDIX A (continued) – Variable Definitions

<i>LitRisk</i>	An indicator variable equal to one if the firm belongs to a high litigation risk industry following Francis, Philbrick, and Schipper (1994). The SICs for the industries are: 2833 through 2836, 3570 through 3577, 3600 through 3674, 5200 through 5961, 7370 through 7374, and 8731 through 8734.
<i>LnAuditorVis</i>	The natural log of the total number of stories published about an auditor's clients during the fiscal year.
<i>LnAssets</i>	The natural log of total assets.
<i>LnBadNewsVis</i>	The natural log of one plus the number of stories a firm appears in during the first two quarters of the fiscal year which have a CSS value indicating negative tone.
<i>LnFee</i>	The natural log of audit fees.
<i>LnFeeRev</i>	The natural log of the total audit fees received by an auditor for the fiscal year.
<i>LnGoodNewsVis</i>	The natural log of one plus the number of stories a firm appears in during the first two quarters of the fiscal year which have a CSS value indicating a positive tone.
<i>LnLagFeeRev</i>	The prior year value of <i>LnFeeRev</i> for an audit firm.
<i>LnMVE</i>	The natural log of the market value of equity.
<i>LnNeutralNewsVis</i>	The natural log of one plus the number of stories a firm appears in during the first two quarters of the fiscal year which have a CSS value indicating neutral tone.
<i>LnVis_EarlyQuarters</i>	The natural log of one plus the number of stories a firm appears in during the first two quarters of the fiscal year.
<i>LnVis_FiscalYear</i>	The natural log of one plus the number of stories a firm appears in during the fiscal year.
<i>LnHighVisRestate</i>	The natural log of the total of the absolute value of net dollars restated by an auditor's client firms in the fiscal year $t-1$ where the client firm was in the top <i>visibility</i> quintile in fiscal year $t-2$.
<i>LnLowVisRestate</i>	The natural log of the total of the absolute value of net dollars restated by an auditor's client firms in the fiscal year $t-1$ where the client firm was in the bottom <i>visibility</i> quintile in fiscal year $t-2$.
<i>LongTenure</i>	Indicator variable set equal to one if an auditor has worked for the client for more than eight years.
<i>Loss</i>	An indicator variable equal to one if income before extraordinary items was negative in the current or either of the two previous years.
<i>MB</i>	The market value of equity divided by the book value of common equity.

APPENDIX A (continued) – Variable Definitions

<i>ModOpinion</i>	An indicator variable equal to one if the firm receives a modified audit opinion which is defined as anything other than a standard unqualified audit opinion coded as one in Compustat.
<i>NewDebt</i>	An indicator variable set equal to one if the firm issues new debt in the subsequent year.
<i>PPE</i>	The lagged value of net property, plant and equipment scaled by lagged total assets.
<i>PrevSent_FY</i>	The mean Composite Sentiment Score (CSS) value for stories used in calculating <i>LnVis_FiscalYear</i> .
<i>Prnocf</i>	An indicator variable set equal to one if the firm had negative operating cash flows in the previous year.
<i>Receivables</i>	Total receivables divided by lagged total assets.
<i>Ret</i>	A firm's two day cumulative return adjusted by the value weighted market return calculated beginning on the day a news story is released and ending the next day.
<i>RetVol_EQ</i>	The natural log of the standard deviation of a firm's daily market adjusted returns for the first two quarters of the fiscal year.
<i>RetVol_FY</i>	The natural log of the standard deviation of a firm's daily market adjusted returns for the fiscal year.
<i>ROA</i>	Operating income after depreciation divided by lagged total assets.
<i>Sales</i>	The natural log of annual revenues.
<i>ShortTenure</i>	Indicator variable set equal to one if an auditor has worked for the client for less than four years.
<i>TradeVol_EQ</i>	The natural log of the mean value of a firm's daily trading volume measured as trading volume scaled by shares outstanding.
<i>TradeVol_FY</i>	The standard deviation of a firm's daily trading volume for the fiscal year where trading volume is calculated as the day's trading volume scaled by the number of shares outstanding.
<i>Turnover</i>	Indicator variable set equal to one if there is an auditor change in the subsequent year.
<i>Year</i>	Fixed effect indicating the observation's fiscal year.

APPENDIX B – RavenPack News Event Groups and Categories

GROUP/category

ACQUISITIONS-MERGERS

acquisition-acquiree
 acquisition-acquirer
 acquisition-completed-acquiree
 acquisition-completed-acquirer
 acquisition-failed-acquiree
 acquisition-failed-acquirer
 acquisition-interest-acquiree
 acquisition-interest-acquirer
 acquisition-merger-termination-fee
 acquisition-opposition-acquiree
 acquisition-opposition-acquirer
 acquisition-regulatory-approval-acquiree
 acquisition-regulatory-approval-acquirer
 acquisition-regulatory-scrutiny-acquiree
 acquisition-regulatory-scrutiny-acquirer
 acquisition-rumor-acquiree
 acquisition-rumor-acquirer
 acquisition-rumor-denied-acquiree
 acquisition-rumor-denied-acquirer
 merger
 merger-blocked
 merger-completed
 merger-failed
 merger-opposition
 merger-regulatory-approval
 merger-regulatory-scrutiny
 merger-rumor
 merger-rumor-denied
 stake-acquiree
 stake-acquirer
 unit-acquisition-acquiree
 unit-acquisition-acquirer

ANALYST-RATINGS

analyst-ratings-change-negative
 analyst-ratings-change-neutral
 analyst-ratings-change-positive
 analyst-ratings-history-negative
 analyst-ratings-history-neutral
 analyst-ratings-history-positive
 analyst-ratings-set-negative
 analyst-ratings-set-neutral
 analyst-ratings-set-positive

ASSETS

asset-sale
 facility-close
 facility-fire-disaster
 facility-open
 facility-sale
 fire-sales
 theft

GROUP/category

BANKRUPTCY

bankruptcy
 bankruptcy-exit
 bankruptcy-fears
 bankruptcy-unit

CORPORATE-RESPONSIBILITY

donation

CREDIT

credit-extension-provider
 credit-extension-recipient
 debt-extension-provider
 debt-extension-recipient
 debt-reduction
 debt-restructuring
 debt-restructuring-approval
 debt-restructuring-complete
 debt-restructuring-considered
 debt-restructuring-failed
 debt-restructuring-rejection
 loan-provider
 loan-recipient
 note-acquisition
 note-sale

CREDIT-RATINGS

credit-rating-action
 credit-rating-affirmation
 credit-rating-confirmation
 credit-rating-corrected
 credit-rating-downgrade
 credit-rating-expected
 credit-rating-matured
 credit-rating-no-rating
 credit-rating-outlook-developing
 credit-rating-outlook-negative
 credit-rating-outlook-positive
 credit-rating-outlook-revision
 credit-rating-outlook-stable
 credit-rating-outlook-unchanged
 credit-rating-paid-in-full
 credit-rating-provisional-rating
 credit-rating-publish
 credit-rating-reinstated
 credit-rating-revision-enhancement
 credit-rating-set
 credit-rating-unchanged
 credit-rating-upgrade
 credit-rating-watch
 credit-rating-watch-developing
 credit-rating-watch-negative
 credit-rating-watch-positive
 credit-rating-watch-removed
 credit-rating-watch-unchanged
 credit-rating-withdrawn-rating

APPENDIX B (continued) – RavenPack News Event Groups and Categories

GROUP/category

DIVIDENDS

dividend
dividend-above-expectations
dividend-below-expectations
dividend-down
dividend-guidance
dividend-guidance-down
dividend-guidance-up
dividend-meet-expectations
dividend-suspended
dividend-up

EARNINGS

earnings
earnings-above-expectations
earnings-below-expectations
earnings-down
earnings-estimate
earnings-estimate-downgrade
earnings-estimate-upgrade
earnings-guidance
earnings-guidance-above-expectations
earnings-guidance-below-expectations
earnings-guidance-down
earnings-guidance-meet-expectations
earnings-guidance-suspended
earnings-guidance-up
earnings-meet-expectations
earnings-negative
earnings-per-share
earnings-per-share-above-expectations
earnings-per-share-below-expectations
earnings-per-share-down
earnings-per-share-meet-expectations
earnings-per-share-negative
earnings-per-share-positive
earnings-per-share-up
earnings-positive
earnings-up

EQUITY-ACTIONS

buybacks
buyback-suspended
capital-increase
fundraising
investment-investor
investment-recipient
ipo
ipo-approved
ipo-completed
ipo-considered
ipo-delayed
ipo-extended
ipo-failed

GROUP/category

EQUITY-ACTIONS

ipo-issuance-decrease
ipo-issuance-increase
ipo-opposed
ipo-price-decrease
ipo-price-increase
ipo-pricing
ipo-rumor
ipo-rumor-denied
ipo-unit
ipo-unit-approved
ipo-unit-completed
ipo-unit-considered
ipo-unit-delayed
ipo-unit-extended
ipo-unit-failed
ipo-unit-rumor
ipo-unit-rumor-denied
ownership-decrease-held
ownership-decrease-owner
ownership-increase-held
ownership-increase-owner
public-offering
reorganization
reorganization-approval
reorganization-complete
reorganization-considered
reorganization-costs
reorganization-delayed
reorganization-denied
reorganization-failed
reorganization-rejection
reorganization-savings
reorganization-unit
reorganization-unit-approval
reorganization-unit-complete
reorganization-unit-rejection
spin-off
stock-splits

INSIDER-TRADING

insider-buy
insider-sell
insider-trading-lawsuit-defendant
insider-trading-lawsuit-plaintiff

LABOR-ISSUES

executive-appointment
executive-compensation
executive-death
executive-firing
executive-health
executive-kidnap
executive-murdered
executive-resignation
executive-salary
executive-scandal
executive-shares-options

APPENDIX B (continued) – RavenPack News Event Groups and Categories

GROUP/category

LABOR-ISSUES

hirings
layoffs
strike
union-pact

LEGAL

copyright-infringement-defendant
copyright-infringement-plaintiff
corruption
defamation-defendant
defamation-plaintiff
discrimination-defendant
discrimination-plaintiff
fraud
legal-issues-defendant
legal-issues-plaintiff
patent-infringement-defendant
patent-infringement-plaintiff
settlement

MARKETING

campaign-ad-release
campaign-ad-retired

ORDER-IMBALANCES

buy-imbalance
delay-imbalance
mkt-close-buy-imbalance
mkt-close-sell-imbalance
mkt-open-buy-imbalance
mkt-open-sell-imbalance
no-imbalance
no-mkt-close-imbalance
sell-imbalance

PARTNERSHIPS

joint-venture
partnership

PRICE-TARGETS

price-target-downgrade
price-target-upgrade

PRODUCTS-SERVICES

award
business-contract
clinical-trials
clinical-trials-complete
clinical-trials-filed
clinical-trials-negative
clinical-trials-positive
clinical-trials-start
clinical-trials-suspended
demand-decrease
demand-increase
patient-enrollment-complete
patient-enrollment-start
patient-enrollment-suspended
product-price-cut
product-price-raise

GROUP/category

PRODUCTS-SERVICES

product-recall
product-release
product-side-effects
project-abandoned
regulatory-product-application
regulatory-product-application-withdrawn
regulatory-product-approval-conditional
regulatory-product-approval-denied
regulatory-product-approval-granted
supply-decrease
supply-increase

REGULATORY

ipo-regulatory-scrutiny
regulatory-product-review-negative
regulatory-product-review-positive
regulatory-product-warning
sec-investigation

REVENUES

revenue-above-expectations
revenue-below-expectations
revenue-down
revenue-guidance
revenue-guidance-above-expectations
revenue-guidance-below-expectations
revenue-guidance-down
revenue-guidance-meet-expectations
revenue-guidance-up
revenue-meet-expectations
revenues
revenue-up

SECURITY

cyber-attacks

STOCK-PRICES

stock-gain
stock-loss

APPENDIX C – RavenPack Sample Observation and Analytic Descriptions

Sample observation of a news story from RavenPack database with relevant news analytics. The headline of the sample observation is: “Ford Adjusts Production to Lower Industry Volume and Shift in Customer Preferences; Profit Outlook Revised.” The database does not reference the actual headline. The actual press release can be viewed at: <http://media.ford.com/pdf/052208BusinessPlanReview.pdf>.

TIMESTAMP_UTC	COMPANY	RELEVANCE	CATEGORY
2008/05/22 13:00:49.047	US/F	100	supply- decrease

ESS	ENS	CSS	WLE	PCM	ECM	RCM	VCM	NIP	NEWS_TYPE
28	100	45	50	0	50	50	50	69	PRESS RELEASE

Variable definitions (from RavenPack’s User’s Manual dated September 6, 2010)

TIMESTAMP_UTC:

The Date/Time (yyyy-mm-dd hh:mm:ss.000) at which the news item was received by RavenPack servers in Coordinated Universal Time (UTC).

COMPANY:

This field includes a company identifier in the format ISO_CODE / TICKER. The ISO_CODE is based on the company’s original country of incorporation and TICKER on a local exchange ticker or symbol.

RELEVANCE:

A score between 0-100 that indicates how strongly related the company is to the underlying news story, with higher values indicating greater relevance. For any news story that mentions a company, RavenPack provides a relevance score. A score of 0 means the company was passively mentioned while a score of 100 means the company was predominant in the news story. Values above 75 are considered significantly relevant.

CATEGORY:

An element or “tag” representing a company-specific news announcement or formal event. Relevant stories about companies are classified into a set of predefined event categories following the RavenPack taxonomy. When applicable, the role played by the company in the story is also detected and tagged.

APPENDIX C (continued) – RavenPack Sample Observation and Analytic Descriptions

ESS (Event sentiment score):

A granular score between 0 and 100 that represents the news sentiment for a given company by measuring various proxies sampled from the news. The score is determined by systematically matching stories typically categorized by financial experts as having short-term positive or negative share price impact. The strength of the score is derived from training sets where financial experts classified company-specific events and agreed these events generally convey positive or negative sentiment and to what degree. Their ratings are encapsulated in an algorithm that generates a score range between 0-100 where higher values indicate more positive sentiment while lower values below 50 show negative sentiment. ESS leverages RavenPack's event detection technology and produces a company specific sentiment score every time an event category is matched. ESS is based on RavenPack's Expert Consensus methodology.

ENS (Event novelty score):

A score between 0 and 100 that represents how "new" or novel a news story is within a 24 hour time window. The first story reporting a categorized event about one or more companies is considered to be the most novel and receives a score of 100. Subsequent stories within the 24 hour time window and about the same event for the same set of companies receive lower scores following a decay function whose values are (100 75 56 42 32 24 18 13 10 8 6 4 3 2 2 1 1 1 1 0...). If a news story is published more than 24 hours after any other similar story, it will again be considered novel and receive a score of 100. The ENS score itself is based on the number of similar stories in the past 24 hour window. This means that an overall chain of similar stories can span more than 24 hours, provided no two stories are more than 24 hours apart.

CSS (Composite sentiment score):

A sentiment score between 0 and 100 that represents the news sentiment of a given story by combining various sentiment analysis techniques. CSS combines the scores of 5 sentiment analytics (WLE, PCM, ECM, RCM, and VCM) using an intuitive set of rules and ensuring no sentiment disagreement amongst the analytics.

WLE (Word and phrase level):

A score that represents the news sentiment of the given news item according to the WLE classifier, which specializes in identifying positive and negative words and phrases in articles about global equities. Scores can take values of 0, 50, or 100 indicating negative, neutral or positive sentiment, respectively.

APPENDIX C (continued) – RavenPack Sample Observation and Analytic Descriptions

PCM (Projections by company):

A score that represents the news sentiment of the given story according to the PCM classifier, which specializes in identifying the sentiment of stories that are only about earnings, developments, and projections news. Scores can take values of 0, 50, or 100 indicating negative, neutral or positive sentiment, respectively. This sentiment score is based on RavenPack's Expert Consensus Methodology.

ECM (Editorials and commentary):

A score that represents the news sentiment of the given story according to the ECM classifier, which specializes in short commentary and editorials on global equity markets. Scores can take values of 0, 50, or 100 indicating negative, neutral or positive sentiment, respectively. This sentiment score is based on RavenPack's Expert Consensus Methodology.

RCM (Reports on corporate actions):

A score that represents the news sentiment of the given news story according to the RCM classifier, which specializes in reports on corporate action announcements. Scores can take values of 0, 50, or 100 indicating negative, neutral or positive sentiment, respectively. This sentiment score is based on RavenPack's Expert Consensus Methodology and has been trained on stories that lead up to a pre-identified corporate action announcement.

VCM (Venture, corporate, mergers and acquisitions):

A score that represents the news sentiment of the given story according to the VCM classifier, which specializes in news stories about mergers, acquisitions and takeovers. Scores can take values of 0, 50, or 100 indicating negative, neutral or positive sentiment, respectively. This sentiment score is based on RavenPack's Expert Consensus Methodology, and has been trained on stories that lead up to a pre-identified mergers, acquisitions and takeover event.

APPENDIX C (continued) – RavenPack Sample Observation and Analytic Descriptions

NIP (News impact projection):

A score taking values between 0 and 100 that represents the confidence that this story will have an impact on the market over the following two hour period. A score of 50 specifically means the classifier was unable to determine if the story had any impact. The training set for this classifier used tick data for a test set of large cap companies and looked at the relative volatility of each stock price measured in the two hours following a similar news flash. The relative volatility is the volatility divided by the mean of volatilities of all companies in the test set during the same period. The classifier learns to predict whether relative volatility is high given the language used by journalists in news flashes, focusing on corporate actions and analyst revisions. Factors like time of day and day of week are taken into account when calculating the confidence that volatility is considered high. The higher the score, the higher the confidence that the story will have an impact in terms of stock volatility. The lower the score, the lower the confidence that the story will have an impact. The best performance of the score is obtained when filtering for company RELEVANCE above 90. This NIP score is based on RavenPack's Market Response Methodology.

NEWS_TYPE:

Classifies the type of story into five news categories: breaking news flash, news flash, full-article, press release, or tabular-material.

Appendix D – News Story Database Examples

Below are examples of a subset of stories for three news events. Examples 1 and 2 each pertain to a separate company. The observations for each cover the same news event category (listed in Appendix B and described in Appendix C) and are linked by a shared ENS_KEY (described in Appendix C). Discussions with RavenPack representatives suggest that the patterns in these examples are fairly typical. Casual readings of a small sample of news stories are consistent with this pattern. Examples 1 and 2 show news flashes being followed quickly by full articles.

NEWS_TYPE	TIMESTAMP_UTC	HEADLINE
Example 1		
HOT-NEWS-FLASH	1/19/2010 2:44	DJ JAL To File For Bankruptcy Protection At Tokyo District Court 0800
FULL-ARTICLE	1/19/2010 3:00	DJ JAL To File For Bankruptcy Protection At 0800 GMT - Sources
FULL-ARTICLE	1/19/2010 3:15	JAL To File For Bankruptcy Protection At 0800 GMT - Sources
HOT-NEWS-FLASH	1/19/2010 7:59	DJ Japan Airlines Files For Bankruptcy Protection
FULL-ARTICLE	1/19/2010 8:24	DJ Japan Airlines Files For Bankruptcy Protection
FULL-ARTICLE	1/19/2010 8:39	Japan Airlines Files For Bankruptcy Protection
FULL-ARTICLE	1/19/2010 10:18	WSJ UPDATE: Japan Airlines Files For Bankruptcy Protection
FULL-ARTICLE	1/19/2010 14:08	WSJ 2nd UPDATE: Japan Airlines Files For Bankruptcy Protection
Example 2		
NEWS-FLASH	2011/01/04 11:01	Quanta Svcs Units Get Pact Valued At About \$118.5M>PWR
FULL-ARTICLE	2011/01/04 11:28	Quanta Services Gets \$118.5M Contract For Fiber Optic Network
FULL-ARTICLE	2011/01/04 11:43	Quanta Services Gets \$118.5 Million Contract For Fiber Optic Network

APPENDIX E – RavenPack Analytics Correlations

Correlations between *CSS* and alternative measures of news story sentiment described in Appendix C. Pearson (Spearman) correlations are above (below) the diagonal. Correlation values are calculated using 4,479,603 news story observations which include all news story types except press releases. This sample includes 1,719,895 news story observations with *ESS* data. Correlations with p-values < 0.0001 appear in bold.

	<i>CSS</i>	<i>WLE</i>	<i>PCM</i>	<i>ECM</i>	<i>RCM</i>	<i>VCM</i>	<i>ESS</i>
<i>CSS</i>	-	0.396	0.456	0.530	0.650	0.627	0.413
<i>WLE</i>	0.456	-	0.268	0.189	0.277	0.165	0.295
<i>PCM</i>	0.437	0.268	-	0.293	0.477	0.293	0.277
<i>ECM</i>	0.846	0.188	0.292	-	0.415	0.266	0.472
<i>RCM</i>	0.629	0.276	0.476	0.414	-	0.418	0.392
<i>VCM</i>	0.344	0.164	0.292	0.265	0.415	-	0.251
<i>ESS</i>	0.544	0.297	0.285	0.504	0.399	0.252	-

APPENDIX F – Company Size/*Visibility* Disparity Examples

Companies are ranked into deciles annually based on size. Companies are also ranked and separated into deciles based on annual press coverage levels. For each company year a size/*visibility* disparity score is calculated as the difference between these two decile rankings. Examples of companies with a disparity score with an absolute value greater than or equal to five are included in (I) and (II) below. Companies listed in (I) have size decile rankings that are greater than their corresponding *visibility* rank by at least five deciles. Companies listed in (II) have size decile rankings that are less than their corresponding *visibility* rank by at least five deciles. Examples of companies with a disparity score of zero are included in (III) and (IV) below. Companies listed in (III) are in the top size decile for both size and *visibility*. Companies listed in (IV) are in the bottom decile for both size and *visibility*.

(I)

Examples of Larger Companies with Lower Levels of Media Visibility

Horace Mann Educators Corp.
Thermo Fisher Scientific Inc.
Gerdau Ameristeel Corp.
WebMD Health Corp.
MediaCom Communications Corp.
Stericycle Inc.

(II)

Examples of Smaller Companies with Higher Levels of Media Visibility

Tivo Inc.
Edgar Online Inc.
AthenaHealth Inc.
Taser International Inc.
Nutrisystem Inc.
Denny's Corp.

*Examples of Companies with no Size/*Visibility* Disparity*

(III)

Large Companies with High Visibility

Bank of America Corp.
JPMorgan Chase & Co.
Exxon Mobil Corp.
Pfizer Inc.
Wal-Mart Stores Inc.
Procter & Gamble

(IV)

Small Companies with Low Visibility

MakeMusic Inc.
Digital Power Corp.
Vita Food Products Inc.
Microwave Filter Co. Inc.
Encision Inc.
Educational Development Corp.

VITA

Michael Penn, Jr. was born in Arizona but moved to Texas before his first birthday. There he learned to walk and talk and to avoid fire ants. He subsequently moved back to Arizona where he relearned to walk and talk and also learned to avoid the fiery rays of the sun. He graduated from Parker High School and attended college for a year at the University of Arizona. He then spent two years as a volunteer fulltime missionary for The Church of Jesus Christ of Latter-day Saints in the Philippines. Upon his return he earned his Bachelor of Science in Business Administration at The University of Arizona's Eller School of Management majoring in Accounting. While completing his degree he worked for the Department of Defense as a contract administrator and contract auditor. He continued working with the Department of Defense following graduation until he returned to school to earn a Master of Accounting degree at The University of Arizona. Following the advice of his wife—who informed him he was more interesting when he was in school than when he was working—he continued his formal education entering the Accounting Ph.D. program at The Pennsylvania State University. Now a Smeal alumnus, Michael is an assistant professor of accounting at The Florida State University. He resides in Tallahassee, Florida with his wife Mikah and their three children (Leora, Audrey, and Charlotte) each of whom Michael hopes will never cross a fire ant.